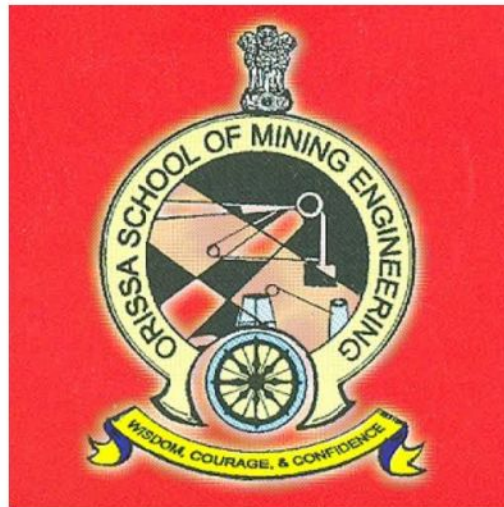


CLASS NOTE
OF
ELECTRICAL ENGINEERING MATERIAL
3rd Semester Electrical Engineering



Orissa School of Mining Engineering, Keonjhar-758001

Prepared By:

Er. Suchismita Sahoo, Lecturer , Electrical Engineering

CHAPTER 1

CONDUCTING MATERIAL

1.0 INTRODUCTION

Since the earliest days of the evolution of mankind, the main distinguishing feature between human beings and other mammals has been the ability to use and develop materials to satisfy human needs. Everything (cars, computers, televisions, medical devices etc) we see and use is made of materials. Materials play a vital role in this modern age of science and technology. Now a days various kinds of materials are used in industries, housing, agriculture, transportation, entertainment, medical care, defense etc to meet our individual requirements.

Materials are evolving faster today than at any time in history. New and improved materials can stimulate innovation and product improvement. History is measured by innovations in materials. The inventions of new materials have been central to the growth, prosperity, security and quality of human life since the beginning of history. The new materials always open the door to new technologies, whether they are in civil, chemical, nuclear, agricultural, mechanical, biomedical or electrical engineering.

1.2 RESISTIVITY

Resistivity : It is defined as the resistance between opposite faces of a unit cube of that material. It is also known as specific resistance. It is denoted by the symbol ρ and is measured in ohm-meter ($\Omega\text{-m}$).

Resistivity depends on the nature of the conductor material. The resistivity of a material is not constant. The most important factor that affects the value of resistivity is the temperature. The resistance (resistivity) of most of the conducting materials increases with temperature. The designer of electrical equipment should know the value of resistivity to determine conducting properties and power loss ($I^2 R$).

Alloying is another factor that affects the resistivity of a material. By adding some impurities (a small percentage of some other material) to a metal its resistivity can be increased. Alloys have higher resistivity than a pure metal. When a metal is alloyed it also acquires higher mechanical strength which is needed for certain applications.

For example, when Copper (60 %) is alloyed with Zinc (40 %), it is called brass. By alloying copper with zinc the resistivity is increased about 4 times, but the tensile strength of brass is much more than that of copper and therefore may be used for making products such as rods, shafts, heavy plates, socket outlets, knife switches etc. Resistivity: It is defined as the resistance between opposite faces of a unit cube of that material. It is also known as specific resistance. It is denoted by the symbol ρ and is measured in ohm-meter ($\Omega\text{-m}$).

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1.3 CLASSIFICATION OF CONDUCTING MATERIALS

The resistivity of a conducting material is not constant and it depends on the nature of the material. On the basis of resistivity, conducting materials are broadly classified into two categories namely,

1. Low resistivity materials and 2. High resistivity materials.

1. LOW RESISTIVITY MATERIALS

possessing very low value of resistivity. These are used for such applications where power loss and voltage drop should be low. These are used in house wiring as conductors for supplying power to various electrical appliances like Television, Fan, Light, Refrigerator etc. These are also used in electrical machines (generators and motors) as windings to carry electric current.

Copper, aluminium, gold, silver, steel, zinc, tin, nickel are examples of low resistivity materials. Some of the copper alloys such as brass, bronze, cadmium are also low resistivity conducting materials. Copper and silver are most commonly used materials. Silver has lower resistivity than copper but due to its high cost its applications are limited. Low resistivity materials besides possessing very low value of resistivity, should also possess the following additional requirements.

Requirements of Low Resistivity Materials: The low resistivity materials should have,

1. *Low temperature co-efficient.*
2. *High mechanical strength.*
3. *Enough ductility and malleability to draw into thin wires.* 4. *Minimum contact resistance for easy soldering.*
5. *High resistance to corrosion.*
6. *Less power loss and voltage drop*
7. *Long service life.*
8. *Low cost and easily available in the market.*

HIGH RESISTIVITY MATERIALS

High resistivity materials possessing high value of resistivity. These are used for such applications where large value of resistance is required. These are used for making resistance elements for heating devices for converting electrical energy into heat energy. These are also used for making rheostats to controlling current, as meter elements such as ammeter shunts and voltmeter multiplier, starters for electric motors, filaments for incandescent lamps.

In general high resistivity materials are alloys of different materials. Manganin, constantan, nichrome, tungsten, carbon, platinum, mercury, eureka are some of the examples of high resistivity materials. High resistivity materials besides possessing high value of resistivity, should also possess the following additional requirements.

Requirements of High Resistivity Materials: The high resistivity materials should have,

1. *Low temperature co-efficient.*
2. *High melting point to withstand high temperature.*
3. *No tendency to oxidize at higher temperature.*
4. *Enough ductility and malleability to draw into thin wires.* 5. *High mechanical strength.*
6. *Long service life.*
7. *Low cost and easy for transportation.*

1.4 PROPERTIES AND APPLICATIONS OF LOW RESISTIVITY MATERIALS

Low resistivity materials are used in house wiring as conductor, for power transmission and distribution, in the winding of motors, generators and transformers. In fact low

resistivity materials are used in all such applications where power loss and voltage drop should be low. The most commonly used low resistivity materials are copper and aluminium

1.4.1 COPPER

It is most widely used conducting material because of its high conductivity (ie. low resistivity). It is a non-magnetic material and has very good physical, chemical and electrical properties. Copper is available in two forms namely:

1. Annealed copper and

2. Hard drawn copper.

The annealed and hard drawn copper are both chemically same but have different mechanical strength due to different heat treatment process used for making them and are shown in Fig. 1.2.

1. Annealed Copper Annealed copper is soft, flexible and has less tensile strength than hard drawn copper. Annealing process involves heating at a specific temperature and then cooling Annealed copper has higher conductivity than hard drawn copper. Annealed copper is used where flexibility is required. It is commonly used in house wiring, winding wires for electrical machines and transformers, in making coils for many purposes.



(a) Annealed copper



(b) hard drawn copper

FIGURE 1.2: (a) Annealed and Hard Drawn Copper

2. Hard Drawn Copper: Hard drawn copper is hard and has high tensile strength than annealed copper. Hard drawn copper is obtained by drawing copper bars or rods in cold condition. Hardening of copper can also be done by alloying it with other metals. Hard drawn copper is about 4% less conductivity than annealed copper. Hard drawn copper is used where tensile strength (mechanical strength) is required. It is commonly used for overhead line conductors and high voltage underground cables and bus-bars.

1.4.2 SILVER

It has the highest electrical and heat conductivity of all metals and the lowest contact resistance. Cold working will reduce this conductivity when heated above its boiling point, silver boils off as a green vapour. Silver is soluble in nitric acid and in hot sulphuric acid.

Applications:

- 1. is used for jewellery, silverware, electrical contacts.*
- 2. Silver is used in the making of solder, and brazing alloys and electrical contacts.*
- 3. High capacity silver zinc and silver cadmium batteries. 4. Silver sulphide is used for inlaying in metal work and in self resetting circuit breakers.*
- 5. It can be deposited on glass or metals by chemical deposition.*
- 6. Silver chloride is used as cement for glasses.*

1.4.3 GOLD

Gold is good conductor of heat and electricity, and is unaffected by air, nitric, hydrochloric or sulphuric acids and most reagents. A mixture of one part with three parts hydrochloric acid, aqua regia, dissolves gold. It is also dissolved by solutions of azoimide. Further, it is attacked by sodium, and potassium cyanide plus oxygen.

Applications:

- 1. It is used for a radiation control coating for spacecraft.*
- 2. It is used in electronic tubes, as gold-plated grid wire, to give high conductivity and suppressing secondary emissions.*
- 3. Gold powder and gold sheet is used for soldering semiconductors, with gold having a good ability to wet silicon at 371°C (725°F)*
- 4. Gold gallium and gold antimony are used in electronic industry (primarily as wire).*
- 5. It is also used in electronic circuit manufacturing.*

1.4.4 ALUMINIUM

Aluminium is silvery in colour. Its electrical conductivity is next to that of copper. It is much lighter (0.48 times) than copper and its resistivity is 1.6 times higher than copper. It is soft metal but when alloyed with some other materials like magnesium, silicon or iron it acquires higher mechanical strength. The disadvantage of aluminium is that, it has high contact resistance and hence it is difficult to solder. The current carrying capacity is 75 % that of copper and its weight is 1/3 times that of copper. For a given resistance the cross-section of the aluminium conductor is 60 % greater

than that of copper and its weight is only 48.3 % of that of copper conductor. Aluminium is used to make conductors, tubes, sheets etc. as shown in Fig. 1.3.



(a) Aluminium Conductor. (b) aluminium tubes

APPLICATIONS OF ALUMINIUM:

Aluminium is widely available in India and is used extensively in the field of electrical engineering. The following are the applications (uses) of Aluminium.

1. It is used in overhead transmission lines for transmission and distribution of power
2. It is used for conductors in flexible wires which are used in house wiring.
3. It is used for making bus-bars.
4. It is used in the form of A.C.S.R (Aluminium Conductor Steel Reinforce) conductors in overhead transmission lines.
5. It is used for making rotor bars in squirrel cage induction motor.
6. It is used in aircraft industry due to its light in weight.
7. It is used for windings of electrical machines and transformers.
8. It forms useful alloys with Iron, copper, zinc and other metals.

1.4.5 STEEL

Steel are alloys of iron and carbon widely used in construction and other application because of their high tensile strengths and low costs. Carbon, other elements and inclusions within iron act as hardening agents that prevent the movement of dislocation that otherwise occur in the crystal lattices of iron atoms.

Electrical steel is an alloy which may have from zero to 6.5% silicon (Si). 5FeSi commercial alloys usually have silicon content up to 3.2% (higher concentration usually provoke brittleness during cold rolling). Manganese and aluminum can be added up to 0.5%

Applications

- 1. Steels are alloys of iron and carbon, widely used in construction and other applications because of their tensile strengths and low costs.*
- 2 They are used in the general engineering and manufacturing sector, but the bulk goes to transport vehicles.*
- 3. All segments of the energy sector, including nuclear, wind power, electric natural gas, demand steel for infrastructure.*
- 4. About 75% of the weight of typical household appliances, comes from steel and Steel is found in appliances like fridges, washing machines, ovens, microwaves sinks, culterly etc..*
- 5 Laminated steel is used in manufacture of transformer.*
- 6. Laminated steel is used in manufacture of rotar and stator of electric motores.*

1.5 STRANDED CONDUCTORS:

Purpose: To reduce the skin and promity effect. In composite conductors sub conductors touch each other. Composite conductors are typically stranded conductors. In composite conductors different elements are used (In ASR conductors aluminium has the properties of light weight, good conductivity and rustlessness and steel has the property of high tensile strength). Composite conductors are employed as they are flexible compared to solid conductor. Composite reduce proximity effect and also reduces skin effect upto certain extent.

As the area of each conductor get reduced there will not be any stray current will also be reduced there by reducing the skin and proximity effect.

1.6 BUNDLED CONDUCTORS:

Purpose: To reduce the corona discharge

Bundled conductors are employed in Extra high voltage (EHV) transmission as al heigher voltages corona effect is sinificant. In bundled sub conductors are placed as certain distance throughout the tranmission lines. This reduces the corona discharge loss and interference with the communiucation lines nearby. As the effective radius of each phase is increased the effective radius get increased thus reducing the corona discharge.

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5. Laminated steel is used in manufacture of transformer.

6. Laminated steel is used in manufacture of rotor and stator of electric motors

1.5 .STRANDED CONDUCTORS:

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1.7 LOW RESISTIVITY COPPER ALLOYS

Alloying is another factor, which affects the resistivity of a material. Alloying is the process in which impurity (a small quantity of some other metal) is added to the base metal. For example brass is an alloy of copper and zinc, where copper is the base metal and zinc is the impurity.

Pure metals like Copper, Aluminium possess low resistivity, any impurity whether metallic or nonmetallic increases their resistivity. The reason for this is that addition of even slightest impurity produces imperfections in the crystal lattice. The influences of various impurities on the resistivity of conductors depends on the metals they contaminate. Alloyed metals acquires properties like higher

mechanical strength which are needed for certain special applications in electrical field.

The effect of metal impurity on the resistivity of a given metal depends on the nature of the alloy formed.

There are 3 types of alloys formed. They are:

(i) Mechanical mixture alloy

(ii) Solid solution alloy

(iii) Chemically combined alloy

1. Mechanical Mixture Alloy When the metals differ widely in atomic volumes and melting point mechanical mixture alloy is formed. When a liquid solution of metals is cooled down, the alloy consists of crystals of both the metals.

2. Solid Solution Alloy: When the atomic volumes of both the metals do not differ by more than 15% solid solution alloy is formed. When such metals are combined together, completely dissolved in each other and form a common crystal lattice, which accommodates the atoms of both the metals. Ex: Copper, Nickel, Alloy.

3. Chemically Combined Alloy When the metals to be alloyed combine chemically, combined alloy is formed Ex: Zinc, Manganese alloy.

Graphical Representation of the Variation of Conductivity for the Following Alloys: 1. Aluminium - Zinc Alloy :

The resistivity in such alloys vary linearly with percentage content of impurity and conductivity vary opposite to that of resistivity as shown in the Fig. 1.18, since conductivity is the reciprocal of resistivity.

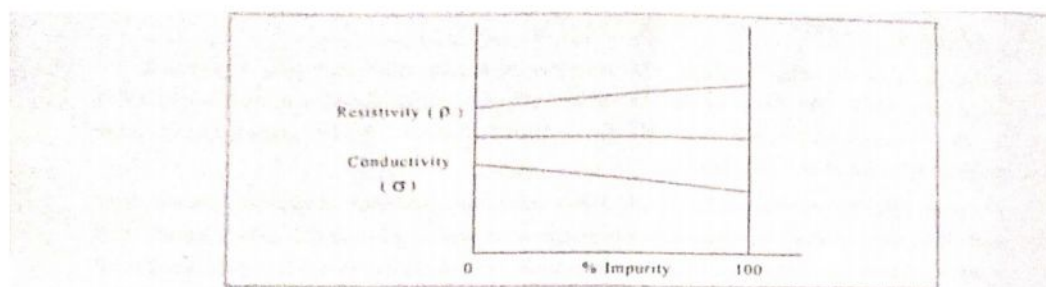
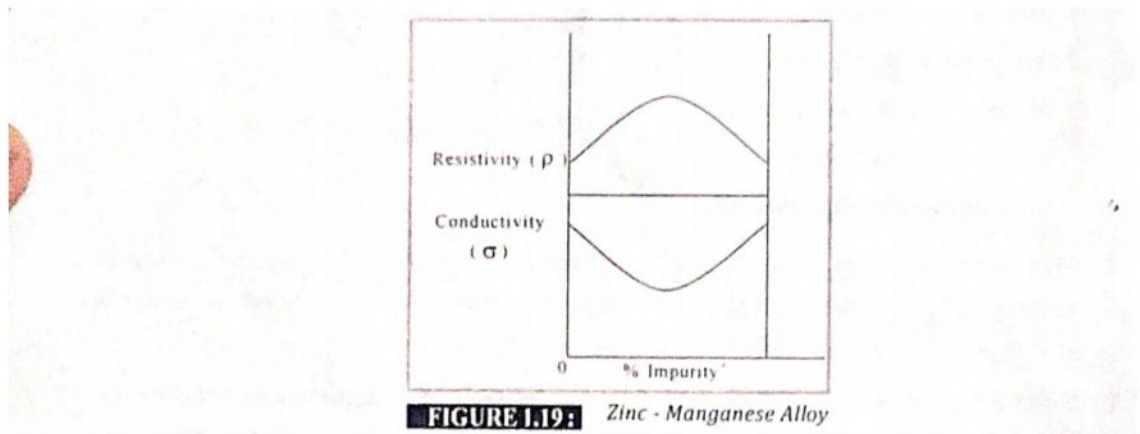


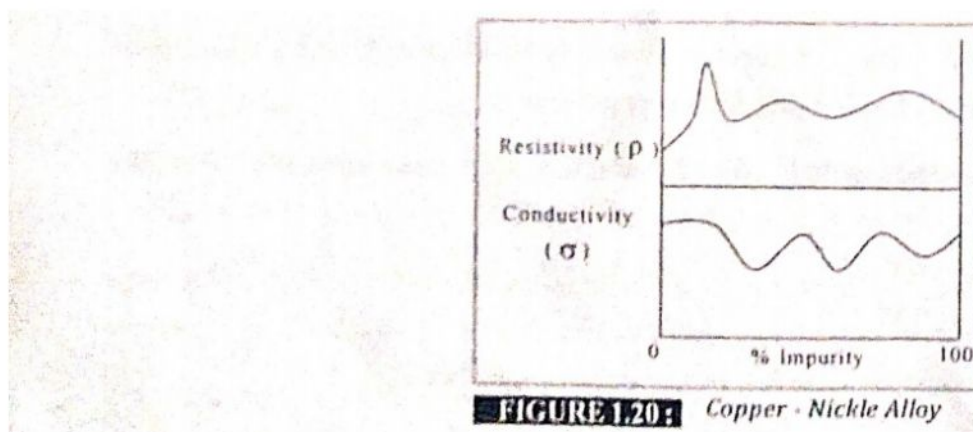
FIGURE 1.18: Aluminium - Zinc Alloy

2. Copper

Nickel Alloy: The resistivity in such alloys increases upto a certain percentage of impurity content and beyond which the resistivity decreases with the increases of impurity content and conductivity varies accordingly as shown in Fig. 1.19



3. Zinc Manganese Alloy: The variation of resistivity and conductivity with impurity content is very irregular (complicated) as shown in Fig. 1.20



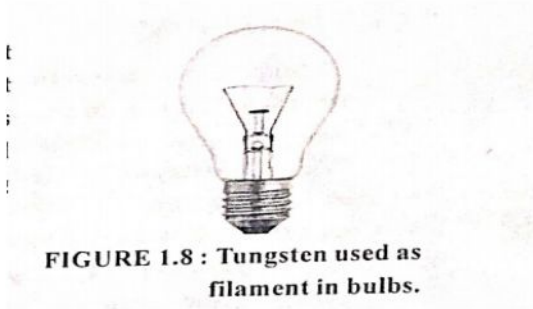
1.8 HIGH RESISTIVITY MATERIALS AND THEIR APPLICATIONS

High resistive materials in general alloy of different materials. Alloys of copper, nickel, chromium, iron, manganese are extensively used as high resistivity materials. These alloys have high resistivity (about 25 times that of pure copper) and very low temperature coefficient of resistance (about 1/20 of that of pure copper). In general higher the resistivity of a material, the shorter, simple and stronger in construction.

High resistivity materials are used for making resistance elements for heating devices, starters for electric motors, coils for electrical measuring instruments, rheostats, filaments for electrical lamps etc. The most commonly used high resistivity materials are Manganin, Constantan, Nichrome, Tungsten, Eureka, Platinum, Mercury and Carbon. The properties and applications of important high resistivity materials are discussed below.

1.8.1 TUNGSTEN

Tungsten is obtained by a suitable process from ore. It does not exhibit magnetic properties when pure but can easily alloy with steel called Tungsten steel which is a magnetic material of top quality. The word "Tungsten" comes from the Swedish language Tung sten. It was identified as a new element in 1781.



APPLICATIONS:

It is used for making:

- 1. Filaments in incandescent lamp due to its high melting point.*
- 2. Electrodes and heater coils in electron tubes to heat the cathode for liberating electrons.*
- 3. An alloy of tungsten steel which is used for making permanent magnets.*
- 4. Electrodes in electron X-ray tubes.*

1.8.2 CARBON

Carbon material is manufactured from graphite (graphite contains 90%) of carbon) and other forms of carbon like coal, coke, charcoal, petroleum etc. The manufacturing process of electrical carbon products involves grinding of the raw carbon material, mixing of the powdered carbon with the binding agent like coal-tar, moulding in the required shape and lastly drying. To increase the conductivity of carbon product, different kinds of additives like copper or bronze powder are mixed with carbon moulding compound. The word "carbon" is derived from Latin word "carbo" (means coal). Without the element carbon, life as we know would not be existing.

APPLICATIONS:

It finds many applications in the form of different kinds of carbon products servicing as current carrying element. The following are the applications of Carbon

ELECTRICAL ENGINEERING MATERIALS

- 1. It is used as a brush in electrical machines.*

2. It is used for electrodes in cells, electric arc furnaces and arc welding.
3. It is used in arc lamps known as carbon arc lamps.
4. It is used for making resistors known as carbon resistors. 5. It is used as filaments in glow lamps known as carbon filament lamps.
6. It is used for anode grids of vacuum valves and tubes.
7. It is used as arcing tips in circuit breakers as they withstand arc effectively.



FIGURE 1.9:Platinum Ore

1.8.3 PLATINUM

Though pure platinum is very good metal, its use is limited due to its high cost and high temperature coefficient of resistance. The word "platinum" is derived from the Spanish term "platina". It will not dissolve in water application. It is used to make jewelry because it will not wear. It is very rare on earth, south Africa is the largest (75%) miner of platinum.

1.8.4 MERCURY

Mercury is the only liquid metal. It is naturally occurring metal and is non-combustible liquid. It has high density.

APPLICATIONS:

It finds many applications in electrical engineering field. The following are the applications of Mercury.

1. It is used in mercury vapour lamps and fluorescent tube lights.
2. It is used in mercury arc rectifiers to convert a.c. into d.c.
3. It is used for making and breaking contact in Buchholz relay which is used for transformer protection.
4. It is used in thermometer for temperature measurements, manometer for measuring pressure of gas in a container, Barometer for measuring atmospheric pressure.
5. It is used as liquid contact material in electrical switches.
6. It is used in paints, coating the back planes of mirrors.

1.9 SUPERCONDUCTIVITY

Superconductivity is the phenomenon where the electrical resistance of a metal disappears when the metal is cooled. Superconductivity occurs in a variety but only when they are cooled to extremely low temperature near absolute zero.

Ever since the discovery in 1911, researchers have sought to raise the temperature at which superconductivity occurs, because that would make possible exciting applications of resistance free electricity long thought impractical in light of the cooling requirements for example supercomputers, frictionless superspeed trains levitated by magnetic force, transmission lines that suffer no power loss through resistance; and supermagnets that could significantly enhance the medical diagnostic technology of magnetic resonance imaging (MRI).

1.10 SUPER CONDUCTING MATERIALS

In 1986, researchers achieved a breakthrough by identifying a new class of high temperature superconducting (HTS) materials. The "high temperature" is high only in the relative sense: these oxide materials remain superconductive above 30 degrees kelvin, which corresponds to 243 degree below zero on the more familiar centigrade scale. However, some HTS materials superconduct at temperature more than 50 degrees higher than the initial application of superconductivity. That makes them easier to cool and easier to use, opening up a wide range of application to take advantages of the superconductor's special properties; zero resistance, magnetic field exclusion, low noise and extremely low power loss for high frequency electronic.

1.11 APPLICATIONS OF SUPER CONDUCTING MATERIALS:

- *Magnetic levitation is an application where superconductors perform extremely well*
- *Superconductors can perform a life saving function in the field of biomagnetic.*
- *Electric generators made with superconducting wire are far more efficient than conventional generators wound with copper wire.*
- *Recently, power utilities have also begun to use superconductors based transformers and "fault limiters".*
- *It is used even at high frequencies, many more filter stages can be employed to achieve a desired frequency response.*

(Theory 4)

Time 3 Hours.

Marks 80

- 1. What is the main effect of temperature on conducting materials.***
- 2. What is ACSR ?***
- 3. What is superconductivity and its application?***
- 4. 5. What is Brass and Bronze? What is Tungsten and its use?***
- 6. Why copper and aluminium are used as current carrying conductor in overhead lines, compare and write their other applications.***
- 7. Explain the effect of temperature on resistivity?***
- 8. What is resistivity?***
- 9. Write the two reasons for using copper in flexible wire? 10. What is the purpose of annealing?***
- 11. Write uses of low resistivity material? 12. Explain properties of high resistive material.***
- 13. Explain the properties and uses of platinum as conducting material.***
- 14. Write two advantages of aluminium over copper.***
- 15. Write the uses of platinum as conducting material***
- 16. What is cunife?***
- 17. Write a short note on A.C.S.R. conductor***
- 18. Write a short notes on carbon as an electric material.***
- 19. Explain properties of high resistive material***

- 20. Write the properties and uses of tungsten?*
- 21. Write a short note on superconductivity?*
- 22. Write two disadvantages of aluminum over copper?*
- 23. Write the uses of silver as conductivity material.*
- 24. Write application of superconducting material.*
- 25. Write the uses of aluminium?*
- 26. What are the advantages of alloying silicon of steel? 27. Write the application for super conducting material.*
- 28. Write a short note on steel?*
- 29. Explain the properties of copper?*
- 30. What is a stranded conductor? 31. Write the uses of mercury as a conducting material?*
- 32. Compare properties of steel and other conducting material.*
- 33. Why is copper used in flexible wire?*

Chapter-2

SEMI CONDUCTING MATERIAL

2.1 INTRODUCTION :

For many years semiconductors played a minor role in the electronics industry. In the year 1948 transistor was invented by the joint efforts of W.Shockley and WHB. Brathan of the Bell telephone laboratory while making a study of the electrical behavior of crystals. Transistors have brought revolution in the electronic industry by using them in computers, satellites, pocket radios, hearing aids. Now a days their application is extended in the building of integrated circuits through which complete circuit functions are possible in a very small piece of semiconductor.

2.2 SEMICONDUCTOR DEVICES :

The branch of engineering which deals with conduction through a vacuum or gas or semiconductor is known as "ELECTRONICS". It creates a great revolution in the field engineering. It essentially deals with electronic devices and their utilization. An electronic device is that in which current flows through a vacuum or gas or semiconductor. These devices are being used in almost all the industries for quality control and automation because such devices have valuable properties and its numerous applications. The electronic devices are capable of performing the function like rectification (ie A.C conservation). Amplification (ie, the process of raising the strength of weak signal), automatic control (i.e, speed of motor, etc.), Generation of D.C. to AC power of any frequency, conservation of electricity into light and vice versa, etc.

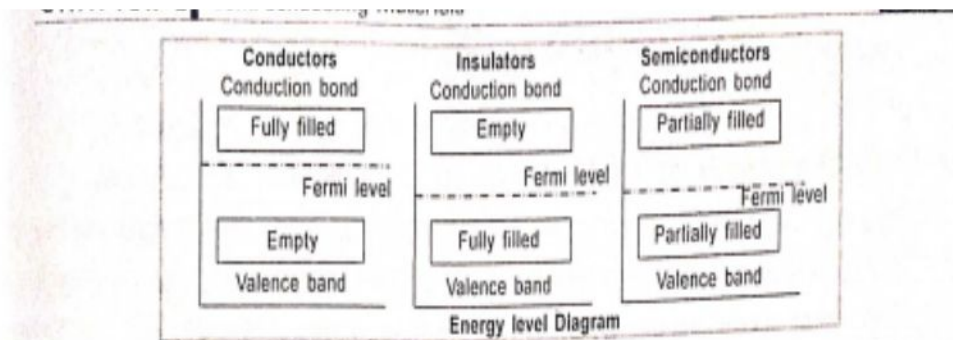
Certain substances like germanium, silicon, carbon etc. are neither good conductors nor insulators. The resistivity of these materials lies in between conductors and insulators. Such substances are classified as semiconductors. In general a semiconductor is defined on the basis of electrical conductivity.

When the outermost orbit of an atom has exactly one half of the maximum of eight electrons i.e., four electrons, the material has both and conductor and insulator properties and is usually a semi conductor.

The resistivity of these materials lies is between ranges of conductor and insulator and is of the order of 10 to $10^{-4}\Omega m$.

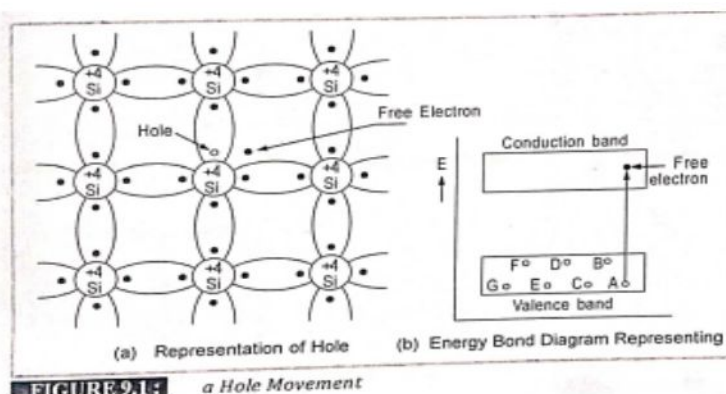
Ex: Germanium, silicon, carbon etc.

Semiconductors have some useful properties and are used in electronic circuits. For instance, transistor is a semiconductor device many other semi conductor devices are becoming increasingly popular. In this chapter we will study the introduction of semiconductors and construction and working of diode and transistor.



HOLE AND ITS MOVEMENT:

At a 0°K temperature the semiconductor crystal behaves as a insulator. However, at room temperature some of the covalent bonds will be broken because of thermal energy supplied to the crystal and conduction is made possible. This is shown in Fig. 9.1, if an electron in the valence band gets sufficient energy, it can jump across the forbidden energy gap and enter the conductor band.



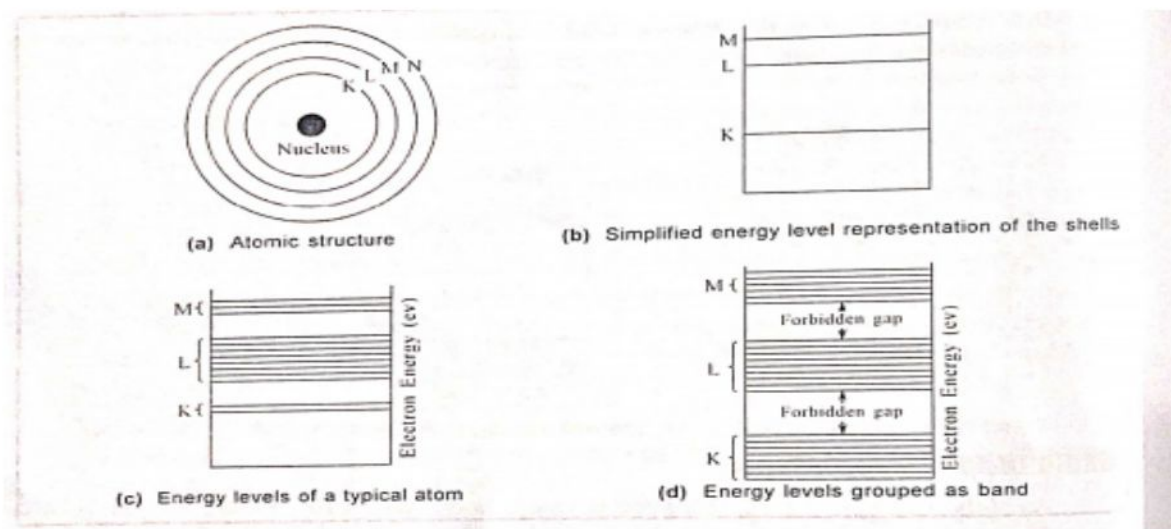
As shown in Fig. 9.1 (a) Suppose covalent bond breaks due to the departure of an electron, a vacancy is left in the valence bond which is called a positive hole.

Suppose the valence electron at A has become free electron due to thermal energy, this creates a hole. The hole is a strong attraction for an electron. A valence electron from nearby covalent bond becomes to fill in the hole at A. This results in a creation of hole at B. In this way by succession of electron movement a hole would appear at G and negative charge would have move from G to A. But it is more convenient to consider a positive hole has moved from A to G.

It should be noted that holes are filled by electrons which move from adjacent atoms without passing through the forbidden energy gap. It means that holes movement takes place in valence bond only movement of hole is less compared to movement of conductor electrons.

2.3 ENERGY BAND THEORY:

The electrons revolve around the nucleus in different orbits or shells as shown in Fig 2.1 (a). The electrons which revolve around the nucleus of an atom has potential energy, centrifugal energy, rotational energy and magnetic energy. These energies together determine the total energy or energy level of an electron and is measured in electron volt (ev). The electrons having minimum energy are in the K-orbit (K-shell) i.e., the orbit closest to the nucleus. Each succeeding level contains electrons with higher energies. It means the larger the orbit in which an electron revolves, the greater is its energy.



If we consider an individual atom, then all the electrons in a given shell possess the same energy as shown in Fig. 2.1 (b). In actual practice, any shell consists of number of electrons. If they revolve in large group, they crowded together. Hence, no two electrons share exactly the same shell. Therefore, the energy of no two electrons is the same. Each shell is therefore divided into sub-shells, each sub-shell having a different energy level. The Fig. 2.1 (c) shows the actual energy levels of an atom, each electron now occupies an energy level different from that of any other. In this figure the energy levels have been grouped into energy bands. Fig 2.1 (d) shows the more general method of representing energy bands. The areas between them are called energy gaps or forbidden energy gap or forbidden gap. In these gaps no electron can have energy.

IMPORTANT ENERGY BANDS IN SOLIDS :

The atom is said to be in an unexcited state when each electron in an atom is in its normal orbit. To move an electron further away from the nucleus, extra energy is required. The extra energy can be

obtained from heat, light, electrostatic, magnetic or kinetic energy. If required amount of energy is absorbed by the electron, it will jump to a higher energy level. When the electron is in the higher energy level, the atom is said to be an excited state.

When required amount of energy is absorbed by the electrons, the energy levels that may be occupied merge into bands of energy levels within any given material. There are three energy bands namely

(i) Valence band

(ii) Conduction band and

(iii) Forbidden energy gap.

(i) Valence Band: The electrons in the outermost binding energy, though their orbital energy is maximum. The electron is called valence band. It is the highest occupied energy level in an atom.

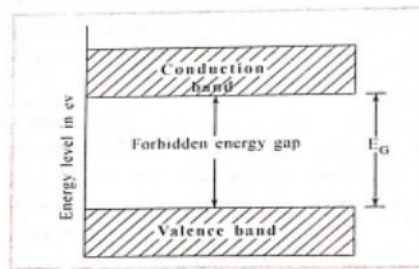


FIGURE 2.2: Energy Band Representation of Ionisation

the least
energy

(ii) Conduction Band: When the required amount of energy is absorbed by valence electrons, it will leave the valence band to become free electron. Such an electron possesses enough energy to be in the conduction band, where it can take part in electric current. These free electrons are responsible for conduction of current. Hence, they are called conduction electrons. The band possessed by conduction electrons is known as "conduction band". It may be either empty or partially filled and is the lowest unfilled energy band in an atom.

(iii) Forbidden Energy Gap: The separation between conduction band and valence band in the energy level diagram is known as "Forbidden energy gap". Normally, electrons do not orbit in the forbidden gap as there is no allowed energy state in this region. The width of this gap is a measure of the bondage of valence electrons to the atom. The greater the forbidden gap more tightly the valence electrons are bound to the nucleus. In order to push an electron from valence band to conduction band external energy equal to the forbidden energy gap must be applied.

2.4 EXCITATION OF ATOMS:

The ground state of the hydrogen atom corresponds to having the atom's single electron in the lowest possible orbit (that is, the spherically "1s" wavefunction, which has the lowest possible quantum numbers). By giving the atom additional energy (for example by the absorption of a photon of an appropriate energy), the electron is able to move into an excited state (one with one more quantum numbers greater than the minimum possible). If the photon has too much energy, the electron will cease to be bound to the atom, and the atom will become ionised.

After excitation the atom may return to the ground state or a low excited, by emitting a photon with a characteristic energy. Emission of photons from atoms in various excited state leads to an electromagnetic spectrum showing a series of characteristic emission lines (including in the case of

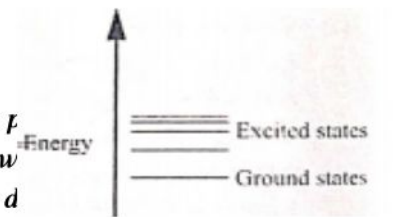
the hydrogen atom, the Lyman, Balmer, Paschen, and Brackets series.)

An atom in a high excited is termed Rydberg atom. A system of highly excited atoms can form a long lived condensed excited state e.g., a condensed phase made completely of excited atoms; Rydberg matter, Hydrogen can also be excited by heat or electricity.

After absorbing energy, an electron may jump from the ground state to a higher excited state.

2.5 CONDUCTORS, INSULATORS AND SEMI-CONDUCTORS :

The electrical properties of a given material depend on the electronic p allowed bands. Electrical conduction is the result of electron motion w electric field is applied to the material, electrons start to move in the d direction of the electric field. An empty energy band (in which there is no free electron) does not of course participate in the formation of an electric current.

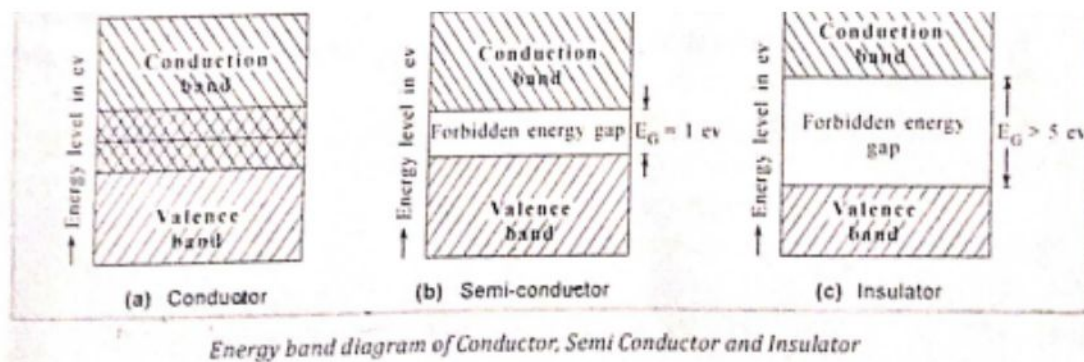


Materials are classified as conductors, insulators, or semiconductors according to their electric conductivity. The classifications can be understood in atomic terms. Electrons In an atom can have only certain well-defined energies, and, depending on their energies, the electrons are said to occupy particular energy levels.

The highest energy band occupied by electrons is the valence band. In a conductor, the valence band is partially filled, and since there are numerous empty levels, the electrons are free to move under the influence of an electric field. Thus, in a metal the valence band is also the conduction band. In an insulator, electrons completely fill the valence band and the gap between it and the conduction band is large. The electrons cannot move under the influence of an electric field unless they are given enough energy to cross the large energy gap to the conduction band. In a semiconductor, the gap to the conduction band is smaller than in an insulator. At room temperature, the valence band is almost completely filled. A few electrons are missing from the valence band because they have acquired enough thermal energy to cross the band gap to the conduction band. As a result, they can move under the influence of an external electric field.

ENERGY LEVEL DIAGRAM OF SEMICONDUCTORS AND INSULATORS :

When the required amount of energy is absorbed by the electron of an atom, it will jump to a higher energy level. The amount of energy required [in electron volt (ev)] to move from one energy level to a higher energy level is different from one material to another material. With in any given material there are two distinct energy bands in which electrons may exist. The two energy bands are valence band and conduction band. These two energy bands are separated by an energy gap called forbidden energy gap in which no electron can normally exists.



The energy gap between the valence band and conduction band is different from material to material. For any given material, the forbidden gap may be large, small or non-existence. The difference between conductor, semi-conductor and Insulator is largely concerned with the relative widths of the forbidden energy gaps.

Fig 31 (a) shows the energy band diagram for conductors. In conductors there is no forbidden gap between the valence band and conduction band. In good conductors valence band conduction bands are overlap with each other. The electrons will move from valence band to conduction even by small amount of energy (about 0.001 eV). By applying a small voltage across such a material a large flow of electrons will result.

Fig 3.1 (b) shows the energy band diagram for semiconductors. In semiconductors a small energy gap existing between valence and conduction bands. Hence, less energy (about 0.2 to 2.5 eV) is required to move electrons from valence band to conduction band. This energy must be more than energy gap E_G . If the energy acquired by the electron is less than E_G , it exists between the two bands. The semiconductors are capable of conducting some electric current even at normal ambient temperature.

Fig 3.1 (c) shown the energy band diagram for insulators. In insulators the forbidden gap between the valence band and the conduction band is quite large. Hence, the electrons require large amount of energy (about 5 to 10 eV) to move from valence band to conduction band. This amount of energy cannot be acquired by the electrons, hence conduction is not possible in insulators. Only at very high temperatures or under high external electric field, an electron can jump from valence band to conduction band. This is called the break down of the insulating material.

2.5 DIFFERENCES BETWEEN CONDUCTOR, INSULATOR AND SEMICONDUCTOR

All materials are made up of atoms. The arrangement of the atoms determines their response to electricity conduction. Like many other things, electrical engineering materials are classified into three groups according to structure, properties or uses namely Conductors, insulators and Semi-conductors. The main difference between these three types of materials is how well they conduct

electricity and how atoms are bound together.

The following are the comparisons between conductors, insulators and semi-conductors.

S.No.	Conductor	Insulator	Semi-conductor
1.	The materials which allow electrical current through them very easily are known as conductors.	The materials which does not allow electrical current through them are known as Insulators.	The materials which allow electrical current partially through them are known as Semi-conductors.
2.	There are more free electrons to jump from one atom to another.	There are no free electrons to jump from one atom to another	There are few free electrons to jump from one atom to another.
3.	Good conductor of electricity.	Bad conductor of electricity.	Partial conductor of electricity.
4.	The resistivity lies between 10^{-8} to 10^{-6} ohm-m.	The resistivity lies between 10^{12} to 10^{18} ohm-m.	The resistivity lies between 10^9 to 10^{12} ohm-m.
5.	Electrical conductivity is very high.	Electrical conductivity is negligible.	Electrical conductivity is very less.
6.	Number of valence electrons are less than four.	Number of valence electrons are more than four.	Number of valence electrons are equal to four.
7.	No forbidden gap between valence band and conduction band ($E_G=0$ ev).	Wider forbidden gap between valence band and conduction band (i.e., $E_G > 5$ ev).	Narrow forbidden gap between valence band and conduction band (i.e., $E_G > 5$ ev).
8.	Very less energy is required for electrons to pass from valence band to conduction band.	Very high energy is required for electrons to pass from valence band to conduction band.	High energy is required for electrons to pass from valence band to conduction band.
9.	They have positive temperature co-efficient of resistance.	They have negative temperature co-efficient of resistance.	They have negative temperature co-efficient of resistance.
10.	Examples are Copper, Aluminium, Tungsten, Silver etc.	Examples are Paper, Wood, Glass, Plastic etc.	Examples are Germanium, Silicon, Selenium, Carbon etc.

exactly equal to four. Another important characteristic of the semiconductor is that they have small energy gap. The energy gap of semiconductors is quite small and varies from 0.2 to 2.5 eV. This property determines the wavelength of radiation which can be emitted or absorbed by the semiconductors and hence helps to construct devices such as light emitting diodes (LED's) and lasers. The semiconductors have negative temperature coefficient of resistance i.e. its resistance decreases with increase in temperature. The best examples of semiconducting materials are Germanium, Silicon, Boron, Selenium, Tellurium etc. Germanium and Silicon are the best known and widely used

semiconductors . There are certain compound semiconductors such as Gallium arsenide, Indium phosphide, Cadmium sulphide etc. The electrical conductivity, energy gap and other properties of semiconductors can be modified by the introduction of impurities (certain other substances like arsenic, antimony, gallium, indium etc.). Such phenomena makes semiconductor devices to use in manufacture of diodes, transistors, SCRS, UITS, Varistors and Thermistors which have many advantages such as compact in size, low cost, light weight, rugged construction etc. It is to be noted that the addition of one part of impurity per million in a silicon material can change that material from a relatively poor conductor to a good conductor of electricity.

INTER ATOMIC BONDS :

The material whether it is a conductor, a semiconductor or an insulator depends largely upon what happens to the outer shell electrons when the atoms bond themselves together to form a solid. The bond between atoms it possible to combine to form a solid. The outermost orbit should have 8 electrons to become stable. The outermost orbit can get some electrons from another atom or give some electrons to another atom to make it 8 and to become stable. This is how atomic bonds are formed. There are three types of inter atomic bonds namely,

- (i) Metallic bond
- (ii) Covalent bond
- (iii) Ionic bond.

(i) Metallic Bond: Elements which have one, two or three valence electrons are very unstable. Atoms of such elements give up their valence electrons to form an electron bond. Metallic bond exists between elements having loosely held valence electrons, so that they can be easily released to the common pool. Solids which held together by this type of bond have properties such as ductility and good electrical conductivity Examples are Silver, Copper, Aluminium.

(ii) Covalent Bond: Covalent bond is formed by sharing of electrons by two or more atoms of same elements. The atoms of materials having 4 or more than 4 valence electrons must share their electrons with the neighboring atoms to have 8 electrons in order to be stable. In covalen bonding each valence of every atom appears to be filled and consequently there are no free electrons within the material. Fig 2.4 shows the

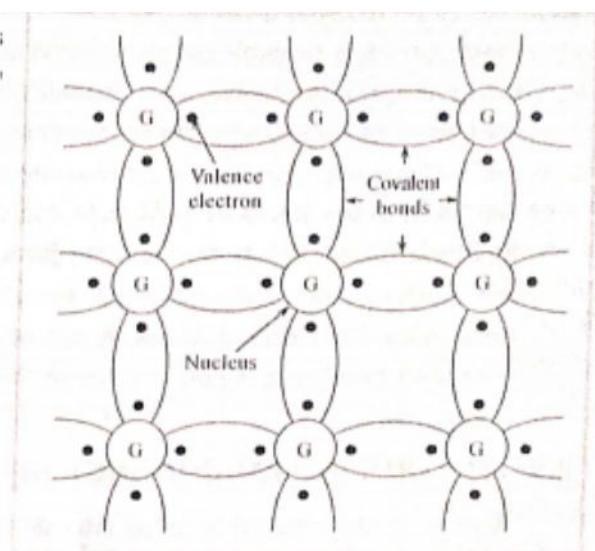


FIGURE 2.4: Covalent Bond of Germanium Atoms

covalent bonding for Germanium in which the atom shares one of its valence electrons with each of the valence electrons of the four Germanium atoms surrounding it.

This sharing of valence electrons produces four pairs of electrons that are covalent bonded and thus completing the total number of electrons in valence shell is 8 to become stable. This bonding provides not only greater strength to such a material but also provides low electrical conductivity because no electrons are available for movement. The electrons are so tightly bond in their orbits and hence require a large amount of energy to remove them from their covalent bonding.

(iii) Ionic Bond: *In ionic bond, atoms of different elements transfer electrons from one to the other, so that both have stable outermost orbits. In this bonding, the atoms which give up electrons become positive ions and the atoms which accept electrons become negative ions. Thus atoms are ionized and hence it is called ionic bonding. This type of bonding occurs in insulating materials.*

For example ionic bond exists between sodium (Na) and Chlorine (Cl). Here, the sodium atom having one valence electron gives up these electrons to Chlorine atom which has seven valence electrons. The chlorine atom accepts these electrons to become stable with eight electrons in the outermost orbit.

2.7 COVALENT BOND:

Only when two atoms of the same elements form a covalent bonds are the shared electron actually shared equally between the atoms. When atoms of different elements

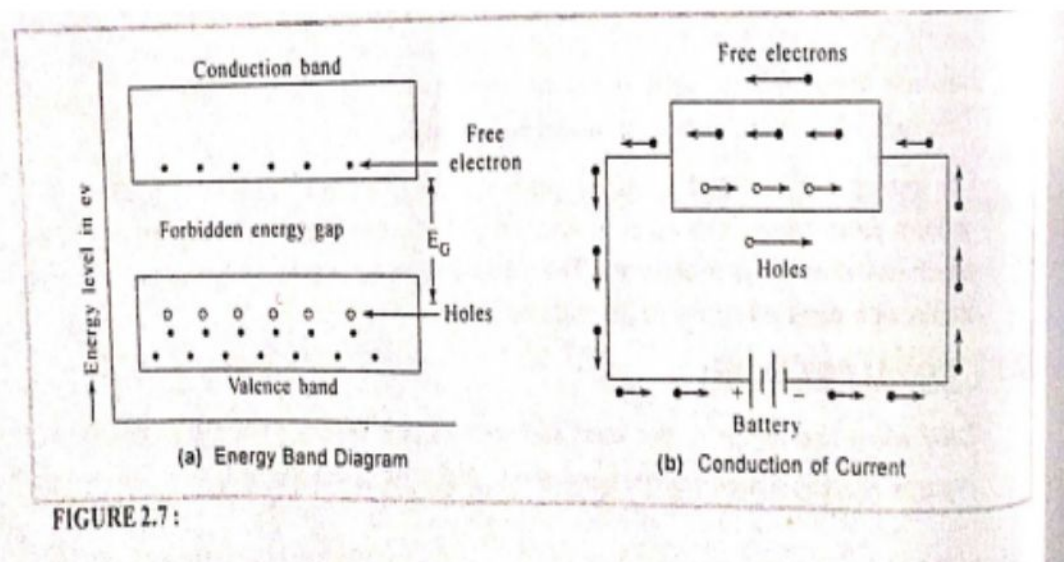
share electrons through covalent bonding, the electron will be drawn more toward the atoms with the higher electronegativity resulting in a polar covalent bond. When compared to ionic compounds, covalent compounds usually have a lower melting and boiling point, and have less of a tendency to dissolve in water. Covalent compounds can be in a gas, liquid or solid state and do not conduct electricity or heat well. The type of covalent bonds can be distinguished by looking at the Lewis dot structure of the molecule. For each molecule, there are different names for pairs of electrons depending if it is shared or not. A pair of electrons that is shared between two atoms is called a bond pair. A pair of electrons that is not shared between two atoms is called a lone pair.

2.8 INTRINSIC SEMICONDUCTOR:

A pure semiconductor is called intrinsic semiconductor. An intrinsic semiconductor is one which does not contain any impurity atoms i.e. it contains only one type of atoms Germanium (Ge) and Silicon (Si) are the two most widely used intrinsic semiconductors because they are in absolutely pure state.

If the temperature is brought down to 0° K (- 273° C), the intrinsic semiconductor will act as a good insulator. If a voltage is applied to it at a temperature above 0° K, it acts as a conductor. The higher the temperature, the more are the free electrons and therefore better is the conduction. When an

electron becomes free from the atom of an intrinsic semiconductor, it breaks a covalent bond and leaves behind a vacancy called a hole as shown in Fig. 2.6 (a). The hole in fact means the loss of an electron. The free electron and the hole form an electron-hole pair. The higher the temperature the greater is the number of electron-hole pairs. The energy required to break the covalent bonds of Germanium is 0.72 eV and for that of silicon is 1.12 eV.



When a voltage is applied to intrinsic semiconductor conduction takes place by electrons and holes. The free electrons move from negative terminal of the voltage source through the semiconductor to the positive terminal of the voltage source. The holes created by the free electrons do not actually move. However, they appear to move from the positive to the negative terminals as shown in Fig 2.6 (b). This is because when a hole is created by an electron breaking a covalent bond due to thermal energy, a valence electron from a neighboring atom may have just enough energy to break its bond and jump over into this hole, thereby creating a new hole. The higher the temperature of a semiconducting material, the lower its resistance and better its conduction. Thus intrinsic semiconductors have negative temperature coefficient of resistance.

2.9 EXTRINSIC SEMICONDUCTOR

The intrinsic or pure semiconductor has little current conduction capability at room temperature. The electrical conductivity of intrinsic semiconductor can be increased by adding suitable amount of impurity or doping agent to it in the process of crystallization. The process of adding impurity to a semiconductor is called a doping and the doped material is called impurity or extrinsic semiconductor. The purpose of adding impurity is to increase the number of charge carriers (either free electrons or holes) in semiconductor crystal. The impurities are added in very small quantities of the order of one atom per million atoms of pure semiconductor. Germanium and Silicon are tetravalent (have four valence electrons). So the impurity or doping material may be either pentavalent (have five valence electrons) or trivalent (have three valence electrons).

Depending upon the type of impurity added, extrinsic semiconductors are classified into two classes namely.

(i) N-type semiconductors and

(ii) P-type semiconductors.

The N stands for Negative and P stands for Positive.

2.10 FORMATION OF N AND P TYPE SEMICONDUCTORS :

The intrinsic semiconductor is of little use as a semiconductor except as a heat or light sensitive resistance. By doping the number of charge carriers and therefore the conductivity of semiconductor can be increased. The doped material is called the impurity or extrinsic semiconductor. Depending upon the type of impurity added extrinsic semiconductors are classified into two types namely,

(i) N-type semiconductors and

(ii) P-type semiconductors.

N-TYPE SEMICONDUCTOR :

When a small amount of pentavalent impurity is added to a pure semiconductor, it is known as N-type semiconductor. Some of the pentavalent impurities are Arsenic, Antimony, Bismuth, Phosphorous etc. When a pentavalent impurity is added to an intrinsic semiconductor (Germanium or Silicon) only four of its valence electrons lock into the covalent bond formation of the atomic structure. The fifth valence electron of the impurity atom is free to move through the crystal. Since, pentavalent atom donate an extra electron, they are called donor impurities. A semiconductor doped with a donor impurity has excess of electrons in its structure, hence it is known as Negative or N-type semiconductor.

Fig 2.7 illustrates the addition of an atom of Arsenic (As) as an impurity to Germanium atoms. Each Arsenic atom forms covalent bonds with the surrounding four Germanium atoms with the help of four of its five electrons. The fifth electron loosely bound to the Arsenic atom. Hence, it is easily removed from the valence band to the conduction band with the application of small external energy. The energy required to remove this fifth electron from the atom is of the order of 0.01 eV for Germanium and 0.05 eV for Silicon.

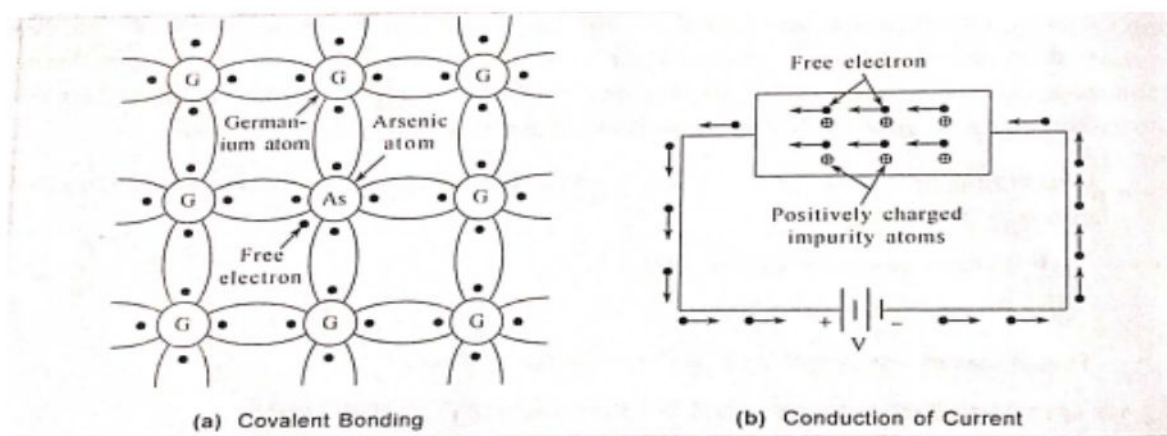


FIGURE 2.7 :

When a voltage is applied to N-type semiconductor, the excess electrons donated by impurity atoms move towards the positive terminal of the battery as shown in Fig. 2.7 (b). This constitutes the electric current. After donation of one electron, the arsenic becomes positive donor ion. In N-type semiconductor the number of electrons are more than holes. The electrons constitute majority carriers while hole constitute minority carriers.

2.11 P-TYPE SEMICONDUCTORS :

When a small amount of trivalent impurity is added to a pure semiconductor, it is known as P-type semiconductor. Some of the trivalent impurities are Boron, Gallium, Indium, Aluminium etc. When a trivalent impurity is added to an intrinsic semiconductor (Germanium or Silicon), these impurity atoms form covalent bonds with four surrounding intrinsic semiconductor atoms but one bond is left incomplete and gives rise to a hole. Since, the impurity has three valence electrons, there is a hole in a covalent bond structure created by the lack of an electron. The hole represents an incomplete covalent bond and exhibits a positive charge. In order to complete the bond and form a stable eight electrons it accepts an electron from a nearby atom. Since trivalent atom accepts an electron, they are called acceptor impurities. Since the crystal is primarily due to holes which have positive charge, such a semiconductor is known as positive or P-type semiconductor.

Fig 2.8 (a) illustrates the addition of an atom of Indium (In) as an impurity to a Germanium atoms. Each Indium atom forms covalent bonds with the surrounding four Germanium atoms but one bond is left incomplete and gives rise a hole. Such impurities have positive carriers because they create holes which can accept electrons. A very small amount of energy is required for an electron to move from valence band to conduction band and holes are created in the valence band by these electrons. These holes constitute the larger number of carriers in the semiconductor material. The energy required to move electrons from valence band to conduction band is of the order of 0.01 eV for Germanium and 0.08 eV for Silicon. The impurity after accepting an electron becomes negatively ionized and the Germanium which released one electron becomes positive ionized.

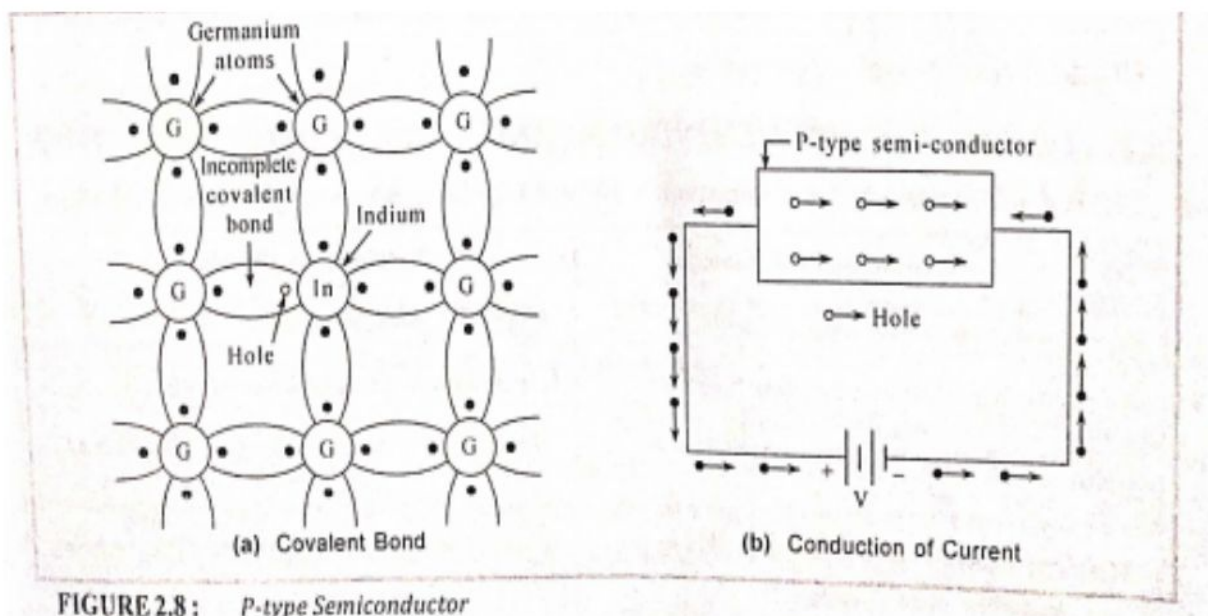


FIGURE 2.8: P-type Semiconductor

When a voltage is applied to P-type semiconductor, the holes are shifted from one covalent bond to another covalent bond. The holes being positively charged move towards negative terminal of the battery and constitute the hole current as shown in Fig 2.8 (b). In P-type semiconductor the number of holes are more than electrons. The holes constitute majority carriers while electrons constitute minority carriers.

FERMI LEVEL

A semiconductor has a energy gap that an electron must overcome to go into conduction. Fermi level is an energy level in the energy gap that represents the point where the probability of electron occupancy is 50 %. It is the level to which electron can fill upto. It allows us to find the density of electrons and holes in a material and hence to determine its electrical properties

For an intrinsic semiconductor, the Fermi level lies in the middle of the energy gap indicating the equal concentration of electrons and holes. For N-type semiconductor, Fermi level is closed to conduction band and for P-type semiconductor it is closer to valence band. The position of Fermi levels are shown in Fig 2.9.

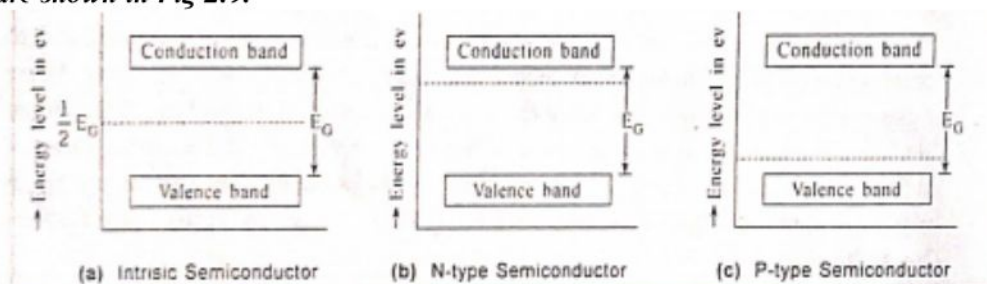


FIGURE 2.9: Position of Fermi Level

2.11.1 DISTINGUISH BETWEEN INTRINSIC AND EXTRINSIC SEMI CONDUCTORS :

The following are the comparisons between intrinsic and extrinsic semiconductors.

S.No	Intrinsic Semiconductor	Extrinsic Semiconductors
1.	It is extremely pure semiconductor.	It is impure semiconductor.
2.	It has poor current conduction capability	It has high current conduction capability.
3.	Number of electrons are always equal to number of holes.	Number of electrons are not equal to number of holes.
4.	Conduction takes place by thermally or optically excited electrons.	Conduction takes place by free electrons or holes.

2.11.2 DISTINGUISH BETWEEN N AND P TYPE SEMICONDUCTORS :

The following are the comparisons between N and P type semiconductors.

S.No	N-Type Semiconductor	P-type Semiconductor
1.	Pentavalent impurity is added to pure semiconductor to form N-type semiconductor.	Trivalent impurity is added to pure semiconductor to form P-type semiconductor.
2.	The impurity atom has one extra electron i.e. five.	The impurity atom has one short fall electron i.e. three
3.	The doping agent is called donor.	The doping agent is called acceptor.
4.	Large number of electrons are produced by doping.	Large number of holes are produced by doping.
5.	In this electrons are the majority carriers and holes are minority carriers.	In this holes are majority carriers and electrons are minority carriers.
6.	Current conduction is predominantly by electrons.	Current conduction is predominantly by holes.
7.	The impurity atom becomes positively ionized by donating an electron.	The impurity atom becomes negatively ionized by accepting an electron.
8.	Examples are Arsenic, Antimony, Bismuth, Phosphorous etc.	Examples are Boron, Gallium, Indium, Aluminium etc.
9.	Fermi level shifts towards conduction band.	Fermi level shifts towards valence band.

2.12 MINORITY AND MAJORITY CARRIERS :

In semiconductors (the materials used to make electronic components like transistors and integrated circuits), in addition to electrons, the travelling vacancies in the valence - band electron population (called "holes"), act as mobile positive charges and are treated as charge carriers. Electrons and holes are the charge carriers in semiconductors.

The more abundant charge carriers are called majority carriers, which are primarily responsible for current transport in a piece of semiconductor. In n type semiconductors they are electrons, which in p type semiconductors they are holes. The less abundant charge carriers are called minority carriers, in n-type semiconductors they are holes, while in p-types while in p-type semiconductors they are electrons.

In an intrinsic semiconductors, which does not contain any impurity the concentration of both types of carriers are ideally equal. If an intrinsic semiconductor is doped with a donor impurity then

the majority carriers are electrons; if the semiconductors is doped with an acceptor impurity then the majority carriers are holes.

2.13 SEMI CONDUCTING MATERIALS :

Semi-conducting Materials: A semi-conductor is a material which has electrical properties between those of insulator and conductors. The electrical conductivity of a semi-conductor is between that of a good conductor and that of an insulator. Semi- conductor is neither a good conductor nor a good insulator. The resistivity is of the order of 100 to $10^2 \Omega\text{-m}$. These are used for the manufacturing of electronic parts such as diodes, transistors, computer chips, photovoltaic cells etc.

Ex: Germanium, Silicon, Selenium, Boron, Uranium, Thorium etc.

2.14 APPLICATION OF SEMICONDUCTORS MATERIALS :

The following are the application of different semi conducting materials.

2.14.1 RECTIFIERS :

A rectifier is an electrical device that converts alternating current (AC) which periodically reverses direction to direct current (DC), which flows in only one direction. The process is known as rectification.

Application:

- The primary application of rectifiers is to derive DC power from AC supply.*
- Rectifiers are also used for detection of amplitude modulated radio signals. Rectifiers supply polarised voltage for welding*
- Rectifiers are now being increasingly used for field excitation of rotating machines.*
- Rectifiers are used in manufacturing electronic devices.*
- Rectifiers are also used in mobile chargers.*

2.14.2 TEMPERATURE SENSITIVE RESISTERS OR THERISTORS :

Thermistor is special type of resistor, whose resistance varies more significantly with temperature than in standard resistors. Generally, the resistance increases with the temperature for most of the metals but the thermistors, responds negatively i.e., the resistance of the thermistors decreases with the increaser in temperature. This is the main principle behind thermistor.

Application:

-- PTC thermistors were used as timers in the degaussing coil circuit of most CRT displays, We can also use PTC thermistors as heater in automotive industry to provide additional heat inside cabin with diesel engine or to heat diesel in cold climatic conditions before engine injection.

-- We can use PTC thermistors as current limiting devices for circuit protection, as replacement for fuses.

-- We can also use NTC thermistors to monitor the temperature of an incubator NTC thermistors can be used as inrush current limiting devices in power supply circuits.

-- NTC thermistors are used as resistance thermometers in low temperature measurements of the order of 10K.

-- We can regularly use the thermistors in the hot ends of 3d printers.

2.14.3 PHOTOCONDUCTIVE CELLS :

The photoconductive cell is a two terminal semiconductor device whose terminal resistance will vary (linearly) with the intensity of the incident light. For obvious reasons, it is frequently called a photoresistive device.

Application:

1. Photocells have myriad uses, especially as switches and sensors.
2. They are a common fixture in robotics, where they direct robots to hide in the dark, or to follow a line or beacon.
3. They are used as timers or measure the speeds of runners during a race.
4. Automatic lights that turn on when it gets dark use photocells as well as street lights that switch on and off according to whether it is night or day.
5. Varistors are used as cable TV system surge protectors.
6. Varistors are used as power supply protection.
7. Varistors are used as Microprocessor protection.
8. Varistors are used in Electronics equipment protection.
9. Varistors are used in low voltage board level protection.
10. Varistors are used in Transient voltage surge suppressor (TVSS).
11. Varistors are used in Industrial high energy AC protection.

2.14.6 TRANSISTORS :

A transistor is a semiconductor device used to amplify and switch electronic signals and electrical power. It is composed of semiconductor material with at least three terminals for connection to an external circuit. A voltage or current applied to one pair of the transistor's terminals changes the

current through another pair of terminals. Because the controlled (output) power can be higher than the controlling (input) power, a transistor can amplify a signal. Today, some transistors are packaged individually, but many more are found embedded in integrated circuits.

Application:

1. For switching applications transistor is biased to operate in the saturation or cutoff region. Transistor in cutoff region will act as an open switch whereas in saturation will act as a closed switch.
2. They are used to make logic gates.
3. Some of the most powerful transistor applications involve amplification, turning a low power signal into one of higher power.
4. Transistors are used to amplify voltage, current and power signals.
5. They are also used in microcontrollers, microprocessors, and other integrated circuits.

2.14.7 HALL EFFECT GENERATOR:

When a current-carrying semiconductor is kept in a magnetic field, the charge carriers of the semiconductor experience a force in a direction perpendicular to both the magnetic field and the current. At equilibrium, a voltage appears at the semiconductor edges is called hall effect.

The Hall effect is the production of a voltage difference (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and a magnetic field perpendicular to the current.

Application:

1. Common applications are often found where a robust and contactless switch or potentiometer is required.
2. Hall effect devices used in motion sensing and motion limit switches can offer enhanced reliability in extreme environments.
3. Some types of brushless DC electric motors use Hall effect sensors to detect the position of the rotor and feed that information to the motor controller.
4. Applications for Hall Effect sensing have also expanded to industrial applications, which now use Hall Effect joysticks to control hydraulic valves, replacing the traditional mechanical levers with contactless sensing..
5. These are used in various sensors such as rotating speed sensors (bicycle wheels, gear teeth,

automotive speedometers, electronic ignition systems), fluid flow sensors, current sensors, and pressure sensors.

2.14.8 SOLAR POWER :

Solar energy which is a combination of light and heat is produced by sun. This energy moves from sun and reaches the earth where human collects it through solar collectors and convert it into any desirable form of energy. According to an assumption this renewable source of energy is powerful enough to replace the need of electricity that we get from 650 barrels of oil per year. To convert sunlight into electricity solar panels, photoelectric technologies and thermoelectric technologies are used.

Application:

- 1. Solar thermal technologies can be used for water heating, space heating, space cooling and process heat generation.*
- 2. Solar cookers use sunlight for cooking, drying and pasteurization. They can be grouped into three broad categories: box cookers, panel cookers and reflector cookers.*
- 3. Agriculture and horticulture seek to optimize the capture of solar energy in order to optimize the productivity of plants.*
- 4. Development of a solar-powered car reduces the fuel consumption.*
- 5. Solar hot water systems use sunlight to heat water. In low geographical latitudes (below 40 degrees from 60 to 70% of the domestic hot water use with temperatures up to 60 °C can be provided by solar heating systems.*

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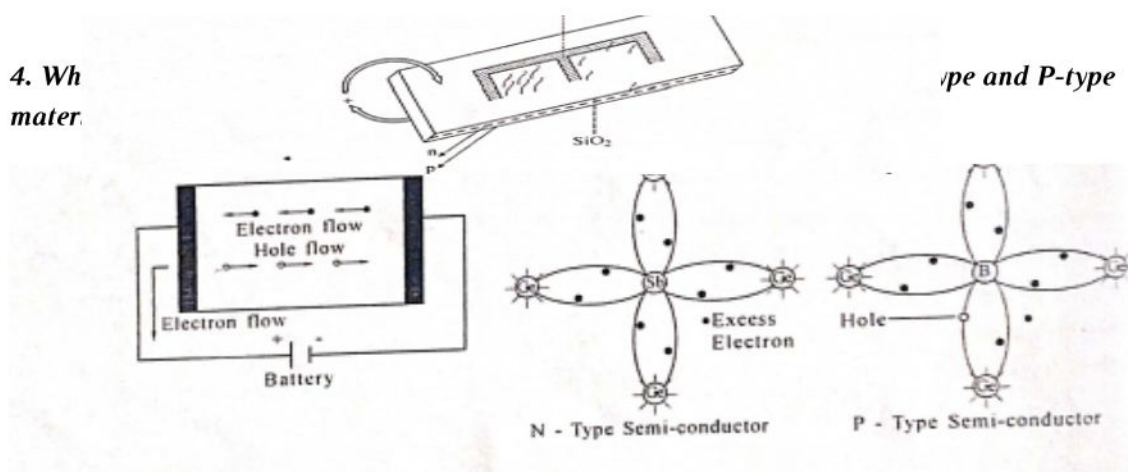
(Theory 4)

Time 3 Hours.

Marks 80

1. Write the uses of varistors?
2. Write names of five numbers of semiconducting material and its use.
3. What is solar cell ?

4. Wh
mater.

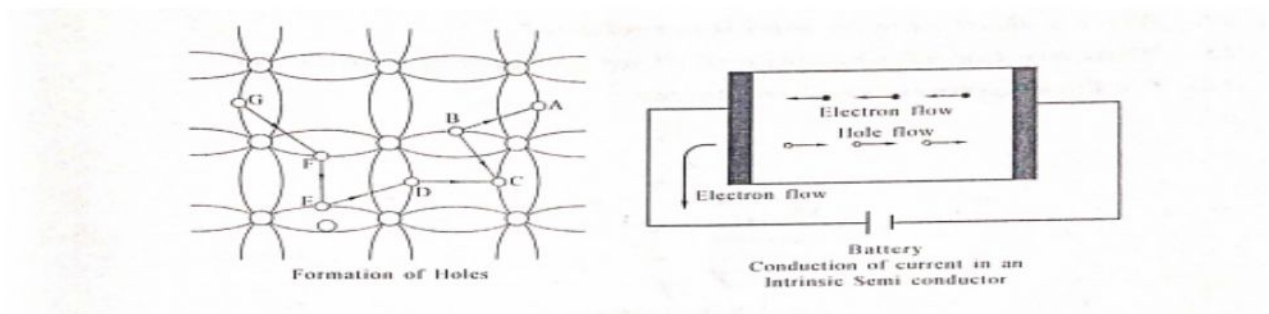


5. What is semiconductor?
6. What are the uses of varister?
7. Write a short note on energy band in solid?
8. Explain function of intrinsic semiconductor?
9. Write a short note on germanium rectifier?
10. Explain classify semiconductors?
11. What is thermister?

12. Explain the energy band diagram of a conductor?

13. Write a short note on silicon rectifier?

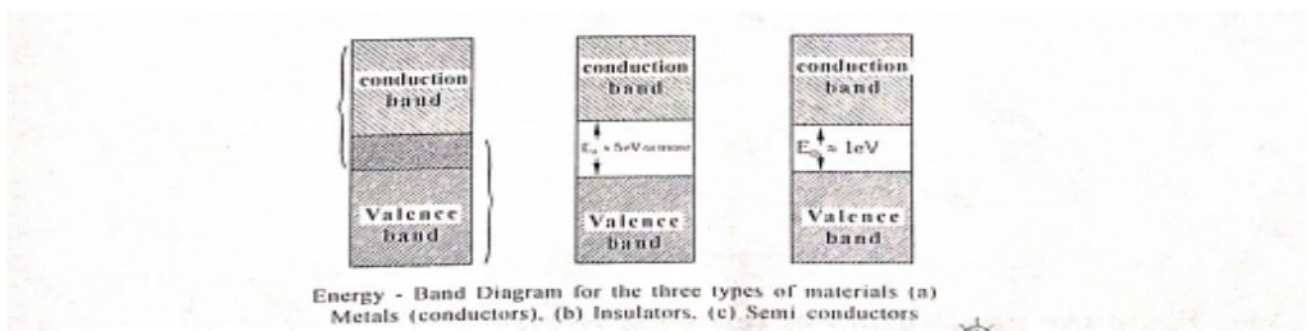
14. Explain conduction in intrinsic semiconductor?



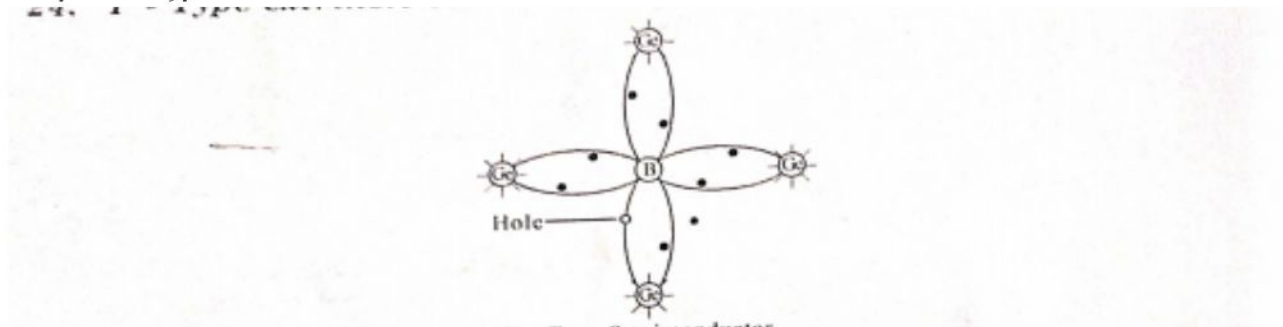
15. What is doping?

16. What are the uses of transistor?

17. Explain the energy band diagram of an insulating material?



18. Explain N-type extrinsic conductor?



N-type Semiconductor

19. Write a short note on copper oxide rectifier?

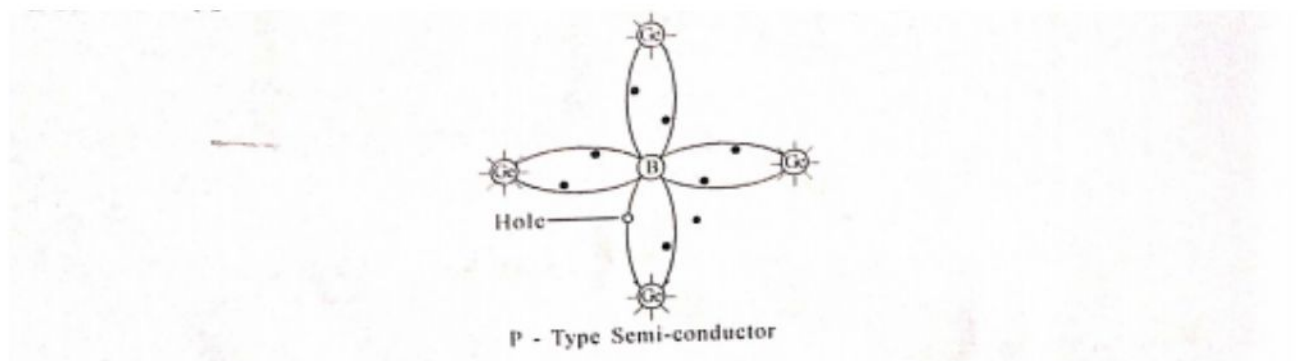
20. Name the element used as semi conductor?

21. Write two applications of photovoltaic cell?

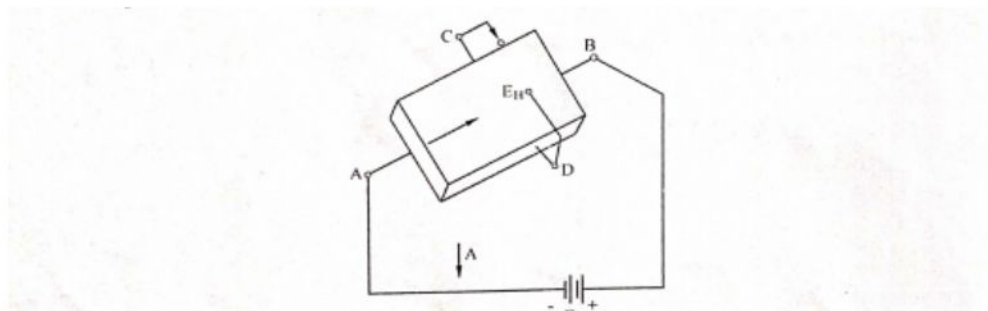
22. Write a short note on selenium rectifier?

23. What are the advantages and disadvantages of transistor?

24. P-Type extrinsic semi conductor



25. Explain hall effect of generator.



26. Write the application of selenium rectifier? 27. What is copper oxide rectifier?

28. Write a short note on extrinsic semiconductor?

29. Write a short note on Photoconductive cell?

30. Explain the characteristics and uses of a transistor?

CHAPTER-3

INSULATING MATERIALS

3.1 INTRODUCTION

Electricity is one of the most common needs in today's modern world. It is however still one of the most dangerous, if it is misused or its applications of use are underestimated. This is the reason why any material that carries electrical current is designed to provide safely with some type of insulation. The insulation limits the flow of current between the different conductors and between the conductors to ground

Insulating materials are non metallic materials that do not allow electric current to pass through it at normal temperatures. There are number of insulating materials used in Electrical apparatus and machinery. They may be organic or inorganic, natural or synthetic. There are many properties such as resistivity, breakdown voltage, thermal stability etc that determine the suitability of a material to be used as insulating material.

An electrical insulator is a material whose internal electric charges do not flow freely. and therefore make it very hard to conduct an electric current under the influence of an electric field. A perfect insulator does not exist, but some materials such as glass, paper and Teflon, which have high resistivity, are very good electrical insulators. A much larger class of materials, even though they may have lower bulk resistivity, are still good enough to insulate electrical wiring and cables.

The insulation is usually made from a non-metallic material. The majority of electrical insulation is made of PVC, plastic, rubber or glass. Many of the best insulators are man-made.

3.2 GENERAL PROPERTIES OF INSULATING MATERIALS

The materials which insulate i.e. prevent the flow of current through it when a potential difference is applied across it is known as "Insulating materials or Insulators". These materials offer high resistivity (10^{12} to 10^{18} Ω -m) to obstruct the flow of electric current without any appreciable power loss. The function of insulating material is to prevent any leakage of current between conductors or conductor to earth. They are available in solids, liquids and gases and almost all insulating materials have negative temperature coefficient of resistance i.e., their resistance decreases with increase in temperature. Hundreds of insulating materials are available in the market.

The selection of right insulating material for a particular purpose is very important because the life of the electrical equipment depends on the type of insulating material used. The technique to dig the right material out of an ocean of materials is to know the properties of materials. Insulating materials are used for many purposes in day to day life. In electrical wiring, conductors are covered with plastic, rubber or some kind of varnished cotton, porcelain Insulators are

used in overhead transmission and

distribution lines, mica or asbestos is used in electric iron to prevent leakage current and electric shock to the human beings.

3.2.1 ELECTRICAL PROPERTIES

Electrical Properties: *Insulation resistance, volume resistance, surface resistance, dielectric strength, dielectric loss, power factor etc are the various electrical properties of insulating materials.*

3.2.2 VISUAL PROPERTIES

Visual Properties: *Appearance, colour, smooth surface are some of visual properties towards customer point of view.*

3.2.3 MECHANICAL PROPERTIES

Mechanical Properties: *Viscosity, porosity, solubility, mouldability, density, moisture, absorption, surface tension, uniformity etc are various mechanical properties of insulating materials.*

3.2.4 THERMAL PROPERTIES

Thermal Properties: *Melting point, thermal expansion, thermal conductivity, softening point, heat resistance, thermal stability, flash point etc are some of the thermal properties of insulating materials.*

3.2.5 CHEMICAL PROPERTIES

Chemical Properties: *Chemical resistance, resistance to chemicals in solids, hygroscopicity, effect of contact with other materials, oxidation, ageing are various chemical properties of insulating materials.*

3.2.6 AGEING:

Ageing: *Insulation resistance decreases with age. This is known as ageing of insulator. For example, the life of underground paper insulated cable is 40 to 50 years. If it used more than the life period than the cable gets shorted and there will be damage of other devices.*

3.3 CLASSIFICATION OF INSULATING MATERIALS

There are hundreds of insulating materials are available in the market. The insulating materials can be broadly classified into three groups. They are,

1. Solid Insulating Materials:

- (1) Fibrous materials (wood, card board, cotton, silk, asbestos, paper etc.)*
- (ii) Impregnated fibrous materials*
- (i) Plastic materials (P.V.C.)*
- (iv) Rubber materials*
- (v) Mineral materials (mica & mica products, marble, slate etc)*
- (vi) Ceramic materials (Porcelain, Steatite, Alumina, Titanate etc.)*
- (vii) Glass materials*
- (viii) Non-resinous materials*

2.Liquid Insulating Materials:

- (i) Mineral oils (kerosene oil, alcohols, lubricating oils, waxes etc)*
- (ii) Synthetic liquids*
- (ii) Vegetable oils (obtained from seeds of cotton, sunflower, soybean, pulp etc.)*
- (iv) Fluorinated liquids (similar to synthetic liquids but have excellent cooling properties)*
- (v) Silicon fluids*
- (vi) Vaseline*
- (vii) Varnish*

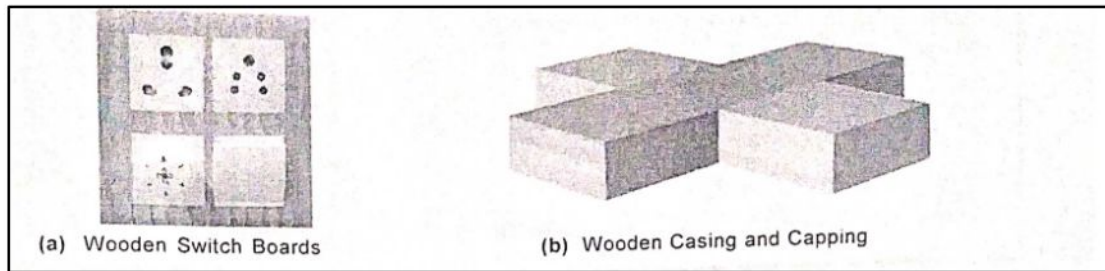
3.Gaseous Materials :

- (i) Air*
- (ii) Nitrogen*
- (iii) Hydrogen*
- (iv) Inert gases (Neon, Argon, Mercury vapour, Sodium vapour)*
- (v) Halogen*
- (vi) Sulphur hexa fluoride (SF₆)*
- (vii) Carbon dioxide (CO₂)*

3.3.2 FIBROUS INSULATING MATERIALS

These are obtained from animal origin or from cellulose which is the main constituent of vegetable plants. They consist of elongated particles called fibres in their structure. In certain materials like cloth, tape, thread etc the cellulose fibres are clearly visible. They are very strong, cheap and hygroscopic. The following are some of the fibrous insulating materials.

1. Wood: *It is naturally available material and obtained from trees. It is good insulating material and is used for low voltages only.*



(a) Wooden Switch Boards

(b) Wooden Casing and Capping

FIGURE 3.2:
Engineering

Applications of wood in Electrical

PROPERTIES:

The following are the properties of wood.

(i) It is light in weight.

(i) It is cheap and easily available in nature.

(iii) Its density is 0.5 to 1.0 gm/cm³. (iv) Its tensile strength varies from 700 to 1300 Kg/cm².

(v) It is hygroscopic.

(vi) Its working temperature is low.

APPLICATIONS:

It is used for making :

(1) Terminal boxes, switch boards, casing and capping, round block in house wiring. (ii) Electrical poles for transmission and distribution of electric power.

(iii) Slot wedges in motors and generator windings.

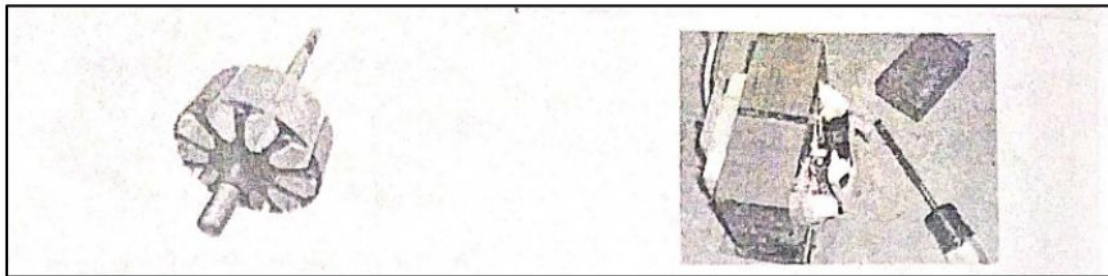
(iv) Instrument and equipment covers.

(v) Handles for tools.

(vi) Separators in batteries.

2. Cardboard: *It is similar to paper but only difference is that its thickness is high. It is made by passing wood pulp through heavy machines called callenders. It is*

used for low voltage only upto 1000 V.



*(a) As Slot Wedges
Windings*

(b) As Separators in Transformer

FIGURE 3.3
Engineering

Applications of Cardboard in Electrical

PROPERTIES:

The following are the properties of Cardboard.

- (i) Its thickness is more than paper.*
- (ii) Its density is higher than paper.*
- (i) It is less flexible.*
- (iv) Its insulation resistance is $10^7 \text{ } \Omega\text{-m}$.*
- (v) It is hygroscopic so need to be impregnated.*
- (vi) Its dielectric strength is 50 KV/mm.*
- (vii) It is strong and tough.*

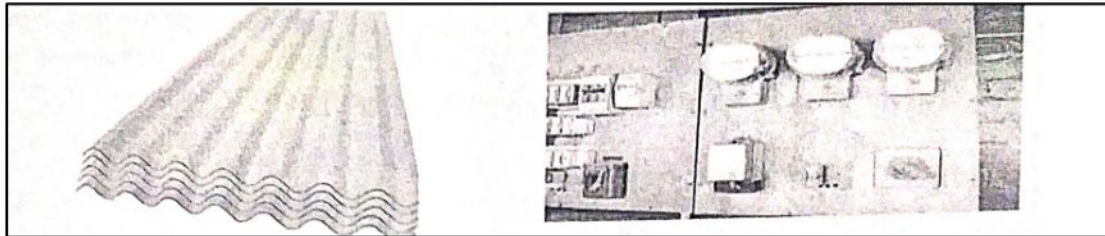
APPLICATIONS:

It is used for making:

- (i) Slot wedges, end liners for starter and rotor core stacks.*
- (1) Separators in transformer windings.*
- (iii) Coil frames in oil immersed transformers.*

3. Asbestos *It is a mineral (naturally occurring) fibrous material. Asbestos materials are impregnated because it absorbs moisture. (Impregnation is the*

process of treating a fibrous material with drying oils, varnishes etc.). Asbestos finds extensive applications in electrical equipment and machines, because it can withstand very high temperature. It is available in kadapa district of Andhra Pradesh. It can be used upto a voltage of 33 KV



(a) As Sheets for house roofs
meters

(b) As Electrical panel boards to fix
meters

FIGURE 3.4:

Applications of Asbestors

PROPERTIES:

The following are the properties of Asbestos.

- 1. It can withstand a temperature of about 400° C. 2. Its dielectric strength is low about 3 to 5 KV/mm.*
- 3. It is hygroscopic hence it is used after impregnation.*
- 4. It is strong and flexible.*
- 5. It has good tensile strength.*

APPLICATIONS:

The following are the applications of Asbestos.

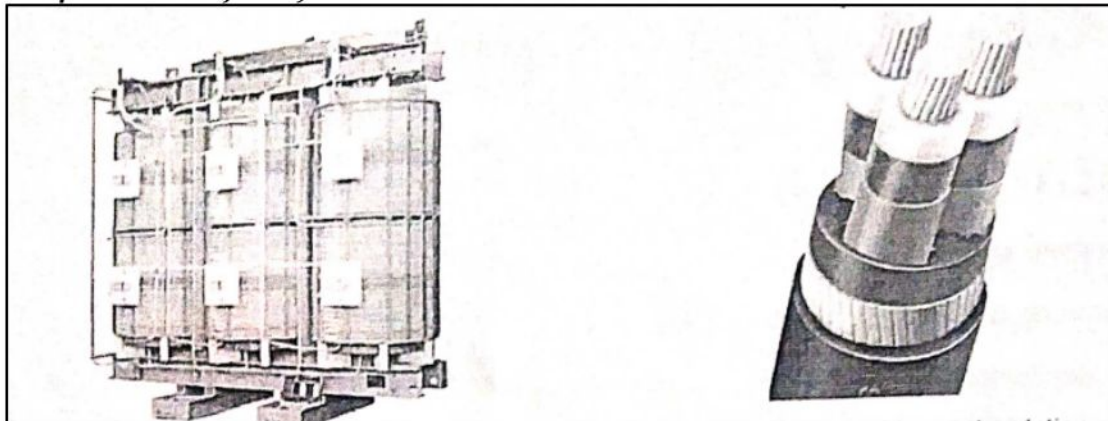
- 1. It is used in the form of boards and sleeves in heating devices like ovens, electric iron etc.*
- 2. It is used as arcing barrier in switchgear and circuit breakers.*
- 3. It is used in coils, windings and insulation in motors and generators.*
- 4. It is used as insulation in transformers.*
- 5. It is used as insulation in wires and cables under high temperature conditions.*
- 6. It is used in panel board construction for switches.*

7. It is used for covering roofs of the buildings.

4. IMPREGNATED FIBROUS INSULATING MATERIAL

In general all the fibrous Insulating materials are impregnated. Impregnation is the process of treating the fibrous material with insulating materials such as resins, bitumens, drying oils, varnishes etc. Impregnation reduces hygroscopicity, chemical and thermal deterioration. It is also helpful in filling up the voids or air spaces in the material. Oils used for this purpose are selected carefully depending on requirement.

1. *Impregnated Paper:* It is prepared from wood pulp or plant fibres by a suitable chemical process. In this process the un-impregnated paper insulation is first heated in vacuum to remove moisture, then impregnated with insulating compounds and finally allowed to cool.



(a) Used in between Transformer Windings as Insulation

(b) Used in Cables as Insulation

FIGURE 3.5: *Applications of Impregnated Paper in Electrical Engineering*

PROPERTIES:

The following are the properties of Impregnated paper.

1. It has good mechanical strength.
2. It can withstand high temperature.
3. It is non-inflammable.
4. It has high dielectric strength (30 KV/mm).
5. It has high insulation resistivity of the order of $10^7 \Omega\text{-m}$.
6. Its dielectric constant is about 3 to 6.
7. It is less hygroscopic hence high durability.
8. The dielectric loss is comparatively less.

9. It is chemically stable.

APPLICATIONS:

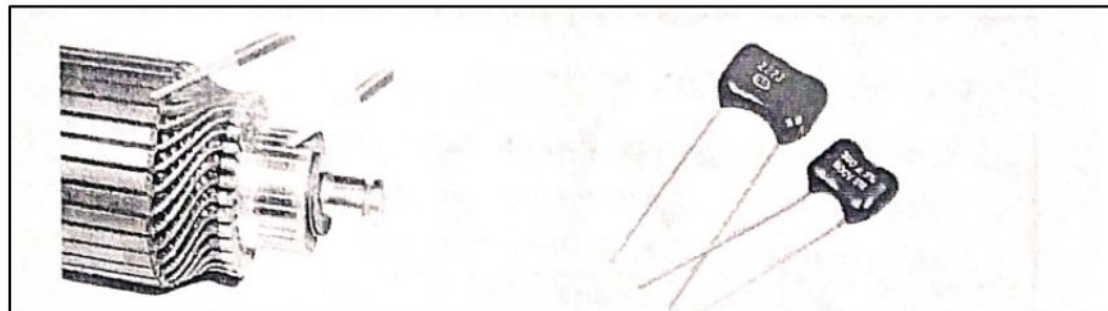
The following are the various applications of impregnated insulating paper.

1. It is used in cables for covering conductors.
2. It is used in high voltage power transformers between windings.
3. It is used for windings in electric machines.
4. It can also be used in the form of tape in jointing the cables.
5. It is used in capacitors.

5. MINERAL INSULATING MATERIALS

The mineral insulating materials are obtained from the mines. These are refined and machined to get the required forms. The most important mineral insulating materials are mica, marble, slate, micanites etc.

1. **Mica:** Mica is an inorganic mineral material obtained from mines. It is one of the best insulating material available in nature. It is a compound of silicate of aluminium with silicate of soda potash and magnesia. Nearly 80% of the total world requirement of mica is found in India. It is available in Nellore district of Andhra Pradesh and Bihar.



(a) Used in between Commutator Segment in Electrical Machines
Dielectric

(b) Used in Capacitor as

FIGURE 3.6:
Engineering

Applications of Mica in Electrical

PROPERTIES:

The following are the properties of mica.

1. It is strong, tough, flexible and rigid material.

- 2. It is non-hygroscopic.*
- 3. It is not affected by acid, alkalies and chemicals.*
- 4. Its dielectric strength is 40 to 150 KV/mm.*
- 5. It can withstand high temperature of about 500° C.*
- 6. It releases water heated at high temperature.*
- 7. It can easily split into thin sheets.*
- 8. It is fire proof.*
- 9. It possess very good insulating properties.*
- 10 Its dielectric constant is 6 to 7.*

APPLICATIONS:

Due to its high insulating properties, the applications of mica are as follows.

- 1. It is used as insulation between commutator segments of d.c. machines.*
- 2. It is used in electrical heating devices such as electric iron, toasters, electric heaters for insulation purpose.*
- 3. It is used in spark plugs of motor vehicles.*
- 4. It is used in capacitors due to its high dielectric constant.*
- 5. It is used in coil frames and slot insulation in electrical machines.*
- 6. Mica in the form of tape is used for taping the stator coils of high voltage alternators.*

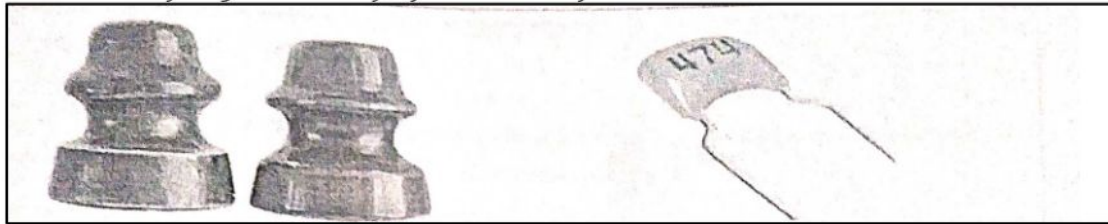
6. CERAMIC INSULATING MATERIALS

Ceramics are clay products which are made by high temperature (about 1500° C) firing treatment. They contain compounds of metallic and non-metallic elements. Structurally ceramics are crystals bonded together. Ceramics used in electrical engineering are Porcelain, Steatite, Alumina, Titanite etc.

PROPERTIES:

The following are the properties of Ceramics.

1. It is mechanically hard and strong.
2. Its dielectric strength is 24 KV/mm.
3. It can be easily moulded to any shape.
4. It is non-hygroscopic.
5. It is not affected by chemical action except by strong acids and alkalies.
6. It can withstand high temperature of about 1000 to 1800° C.
7. It can easily split into thin sheets.
8. Its dielectric constant varies from 5 to 9.5.
9. It has very high resistivity of the order of 10^{11} to 10^{16} Ω -cm at 20° C.



(a) Used as Insulators in Transmission Lines
Dielectric

(b) Used in Capacitors as

FIGURE 3.7: *Applications of Ceramic in Electrical Engineering*

APPLICATIONS:

Due to desirable properties, the ceramic materials are used for many electrical applications. Ceramic insulating materials are used for making:

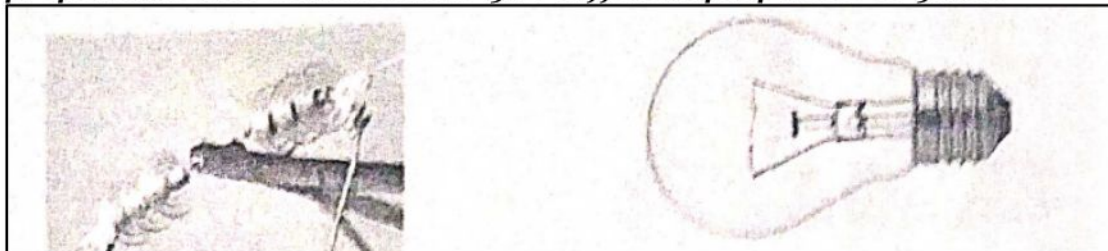
1. Insulators used in transmission and distribution lines.
2. Transformer bushings.
3. Isolators, plugs, sockets and fuse units.
4. Telephone insulators.
5. Heat resistance elements in heating appliances.
6. Discs for electric stoves, electric kettles etc to insulate heating conductors.
7. High temperature appliance like furnace.
8. Equipments for high frequency systems and where thermal shock

resistance is desired.

9. Capacitors and transistors.

7. GLASS INSULATING MATERIALS

Glass is a inorganic material obtained by fusion of different metallic oxides. Different oxides used in the manufacture of glass are sand, silicon, boric acid, sodium, potassium, chalk, magnesium, calcium, lead, kaoline etc. The silicon oxide is most commonly used in the manufacture of glass but the addition of other oxides in different proportions and combinations give different properties to glass.



(a) Used as Insulators in Transmission Lines
Bulbs

(b) Used for Making Electrical

FIGURE 3.8: *Application of Glass in Electrical Engineering*

PROPERTIES:

The following are the properties of glass.

1. It is normally transparent, brittle and hard.
2. It has high dielectric strength of the order of 25 KV/mm to 50 KV/mm.
3. Its resistivity is 10^{10} to $10^{15} \Omega\text{-m}$.
4. Its relative permittivity is 3.8.
5. It has low dielectric loss.
6. It has good mechanical strength.
7. It is resistant to most of the chemicals.
8. It is hygroscopic.
9. It is difficult to manufacture as it requires very high fusion temperature.

APPLICATIONS:

The various applications of glass in electrical engineering are as

follows.

- 1. It is used as dielectric in capacitors.*
- 2. It is used as insulating device in bushings, fuse bodies etc.*
- 3. It is used as line insulators in transmission and distribution lines upto 25 KV.*
- 4. It is used in vacuum tubes, electrical lamps etc.*
- 5. It is used for making scientific instrument parts and containers etc.*
- 6. It is used in chemical industry due to transparency.*

INSULATING RESINS OR PLASTIC INSULATING MATERIALS

Insulating resin or plastic is an organic material and are found in many varieties. It can be moulded into a desired shape during or after manufacture by the applications of heat and pressure. The term plastic is used very frequently in common terminology even for materials which are not really plastics. Most of the plastics are of organic nature composed of hydrogen, oxygen, carbon and nitrogen. Resin is the substitute word normally used in insulation technology in place of plastic. Plastic is available in moulding powders, sheets, rods, tubes, films and liquids. Plastic is light in weight, low cost, water resistant and are available in different colours. Plastics or resins are classified into two types namely.

(i) Natural resins

(i) Synthetic resins.

***1. Natural Resins:** They are available in nature and are derived from plant and animal sources. They are used in preparation of insulating materials. Simple purification or sometimes a little chemical modification is made in natural resins for use as an electrical insulation. The examples of natural resins are Amber, Wood resin, Shellac etc. Wood resins is used for paint and varnish applications and also for the manufacture of mineral insulating oils. Shellac is used for impregnation and in the manufacture of mica.*

***2. Synthetic Resins:** Synthetic resins are made by the chemical reaction of a simple organic compound. These resemble the natural resins in properties such as plasticity and heat resistance, but the chemical composition is widely different from that of natural resins. Now-a-days more than 50% of the insulating materials are synthetic resins.*

These are broadly classified into

(1) Thermoplastic Resins and

(ii) Thermo setting Resins

Thermoplastic Resins

These are formed by polymerization process in which molecules of a compound react together to form new compounds. They can melt and soften when heated and solidify when cooled. The heating cycle can be repeated without any change in the properties of material. These are produced artificially from organic substances like oil and coal products. Fillers like mica, textile, wood, glass fibres are often added in order to get certain desired properties. They are widely used for floor tiles, walls, fluorescent light reflections, lenses etc. Various types of thermoplastics resins are:

- 1. Perspex*
- 2. Polyethylene or polythene*
- 3. Polyvinyl chloride (PVC) etc.*

3.8.2 Thermosetting Resins

These materials undergo a chemical change when moulded. They are hard, rigid and can not be softened by heating to re-shape them. The moulding temperature for these materials is higher than that of thermoplastic materials. When heated first they become soft and on further heating they undergo change and set hard. This process is called thermosetting. It is permanently set and does not soften on application of heat. However, if it is heated further it will breakdown by burning. Fillers like wood, glass fibre, mica, clay, magnesium oxide, aluminium oxide are often added in order to get desired properties. The various types of thermosetting resins are:

- | | | |
|----------------------|--------------------------|-----------|
| <i>1. Bakelite</i> | <i>2. Silicon resins</i> | <i>3.</i> |
| <i>Epoxy resins.</i> | | |

POLYVINYAL CHLORIDE IP.V.C.)

It is a synthetic compound. It is obtained by combination of acetylene and hydrogen chloride in the presence of catalyst (peroxides) at a temperature of about 50° C. It can also be obtained by polymerization of ethylene dichloride and sodium hydroxide in the presence of a catalyst. It does not react with oxygen, oils, alkalies and acids.

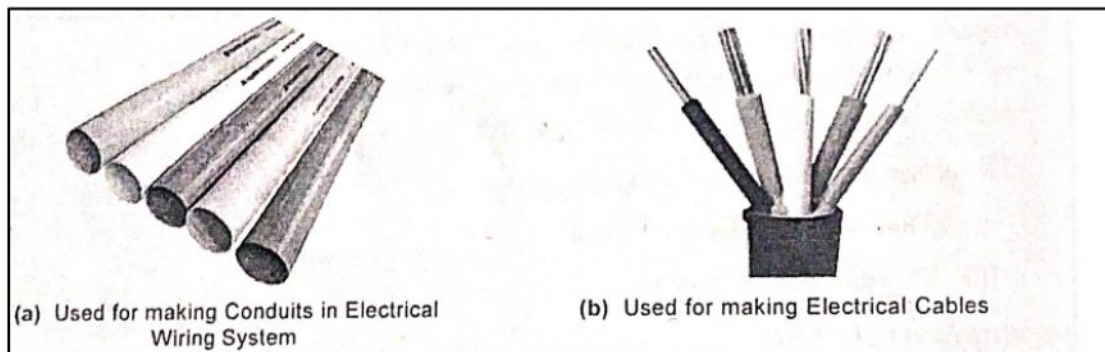


FIGURE 3.9: *Applications of PVC in Electrical Engineering*

PROPERTIES:

The following are the properties of P.V.C.

- 1. It is rigid, hard and little flexible.*
- 2. It never gets corrosion.*
- 3. It is less weight and is cheap.*
- 4. It has high resistance to chemical action.*
- 5. Its insulation resistance is 10^{12} to 10^{13} Ω -cm.*
- 6. It cannot withstand very high temperature (above 80°C).*
- 7. It is normally transparent, brittle and hard.*
- 8. Its dielectric constant is 5 to 6.*
- 9. Its dielectric strength is 15 KV/mm to 30 KV/mm.*
- 10. It is obtained in different colours.*
- 11. It is slightly hygroscopic.*
- 12. It has good electrical and mechanical properties.*

APPLICATIONS:

Due to its good electrical and mechanical properties, it has the following applications.

- 1. It is used in insulation for wires and cables.*
- 2. It is used as insulation for dry batteries.*
- 3. It is used as conduit pipes in electrical wiring system.*
- 4. P.V.C. insulated cables are commonly used for low and medium voltage domestic and industrial light and power installations.*
- 5. It is used as insulation for wires in military, aircraft, communication, radio and television.*

10. ADDITIVES OF P.V.C

The properties of Poly Vinyl Chloride (PVC) can be improved by adding varieties of materials to the basic resin during manufacture. The properties like mechanical strength, porosity, flexibility,

mouldability, moisture, absorption, electrical properties etc can be affected by adding different types of materials.

The properties can be improved by adding the following materials called additives.

1. Fillers

2. Stabilizers

3. Plasticizers

4. Supplementary

additions

1. Fillers Fillers like wood, cotton, asbestos, glass fibres, powdered mica, chalk, quartz etc influence the various properties of P.V.C.

These materials make the P.V.C. cheap.

2. Stabilizers: When P.V.C. is exposed to sunlight or heated to high temperature hydrogen chloride is evolved. This hydrogen chloride affects many properties of the insulating materials and can also corrode the parts of the apparatus which comes in contact with it. Different stabilizers like metal soaps, lead, calcium carbonate, calcium oxide etc are used to prevent the evolution of hydrogen chloride.

3. Plasticizers: These materials reduce the hardness and rigidity of P.V.C and make it flexible and mouldable. Plasticizers like tricresyl phosphate, dibutyl naphthalene control the flexibility of P.V.C.

4. Supplementary Additions: These are used for colouring and lubricating purposes. It is necessary to use different colours to distinguish the different cores of the cables as red, yellow, and blue for the three phases. Different colouring pigments are added to give desired colours, lubricating additives help in moulding process

3.4. GASEOUS INSULATING MATERIALS

Gaseous insulating materials serve two purposes, firstly it acts as an insulant and secondly as heat transfer medium. The gases are most commonly used as insulating materials because they are cheap, easily available, non-inflammable. The dielectric strength plays an important role in the selection of gaseous insulating materials. The dielectric strength of these materials increases with increase in gas pressure, hence they operate under pressure in electrical apparatus. The most commonly used insulating gases are air, nitrogen, hydrogen, neon, argon, sulphur hexafluoride (SF) etc..

Earth's atmosphere is composed of air. Air is a mixture of gases and contains 78% nitrogen, 21 % oxygen with a small quantity of water

vapour, carbon dioxide, argon and various other components.

3.4.2 Commonly used in insulating gas:

Air: It is a naturally and abundantly available gaseous insulator. It needs no processing and can be used directly. It is the most commonly used insulator. It forms about 80% of the earth's atmosphere and under normal conditions it is colourless, tasteless and odorless.

PROPERTIES:

The following are the properties of Air.

1. Its dielectric constant increases linearly with increase in pressure.
2. Its permittivity (ϵ) is 1.
3. Its dielectric strength is 30 KV/cm at 50 Hz.
4. It carries dust and moisture which settle on the parts of windings thereby reduces the efficiency.
5. Oxygen in air corrodes the parts by forming oxides and reduces the life of the equipment.
6. Oxygen in air helps a fire or arc if it arises.
7. It cannot penetrate deep inside the parts from where heat is to be carried out due to its higher density.
8. It carries chemicals, acids, alkalies which damage the insulation and parts of the equipment.
9. It is reliable at low voltages.
10. It has weight, pressure, temperature and can occupy space.

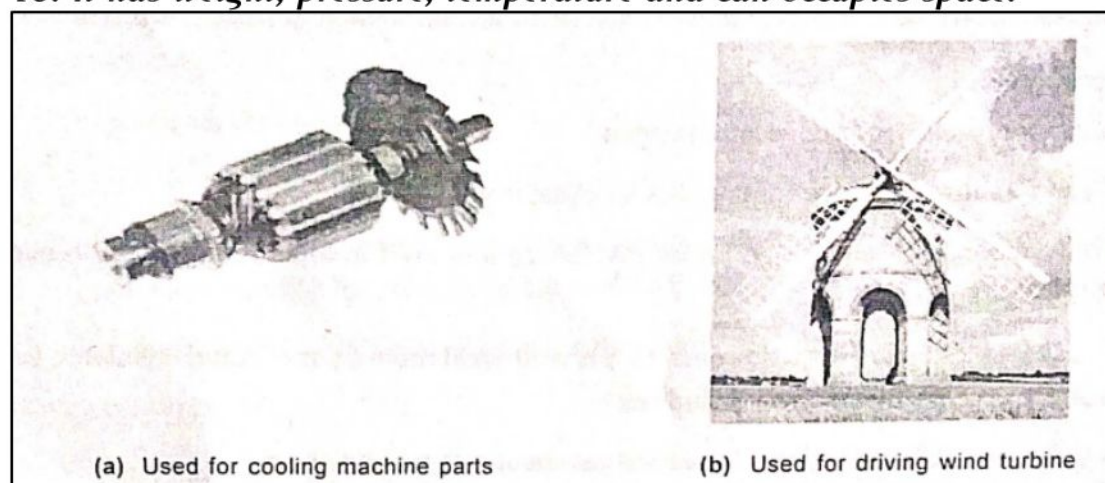


FIGURE 3.10: *Applications of Air in Electrical Engineering*

APPLICATIONS:

Air is necessary for all living things. Some applications of air are as follows.

1. It is used as insulator in overhead transmission lines and air

condensers.

- 2. It is used as an insulator in plugs, switches and various electrical apparatus.*
- 3. Moving air (wind) is used to drive blades of windmills.*
- 4. It is used for cooling parts of machines, transformers and circuit breakers.*
- 5. It is necessary for burning fuels.*
- 6. Compressed air is used for bicycles, scooters, cars, balloons, football etc.*
- 7. Air is needed to hear sound and it carries clouds to different places and helps in bringing rain.*

2. NITROGEN

It is chemically inert. It can be used for both electrical and chemical purpose. It prevents oxidation and thus reduces the rate of deterioration of parts of equipment.

PROPERTIES:

The following are the properties of Nitrogen.

- 1. It is non metal.*
- 2. It prevents oxidation and increases the life of the equipment.*
- 3. It is chemically inert gas..*
- 4. It is used for both electrical and chemical purposes.*
- 5. Its dielectric strength is 3 to 5 V/mm.*

APPLICATIONS:

The following are the applications of Nitrogen.

- 1. It is used as insulator in certain types of capacitors.*
- 2. In conjunction with oil treated paper insulation, it is used in high voltage gas pressure cables.*
- 3. It is used in oil filled transformers to prevent oxidation of the liquid insulator and thereby prevents formation of sludges.*
- 4. It is used as cooling medium in sealed gas insulated transformers.*
- 5. It is used as a safety component in airbags of transport vehicles.*
- 6. Nitrous oxide is used in vehicles to increase the power of engine and speed of the vehicles.*
- 7. It is used to prevent oxidation.*
- 8. It present in all living things including the human body and plants.*

9. Nitric acid (HNO_3) and Ammonia (NH_3) are the compounds of nitrogen which are used in fertilizers.

3. HYDROGEN

It is rarely used as insulator. Electrochemical and electrothermal are employed for generation of hydrogen from water. Electrolyte breakdown of water is a source of hydrogen. At normal temperature and pressure it is a colourless, odorless and tasteless. It reacts with oxygen to form water (H_2O) and hydrogen peroxide (H_2O_2).

PROPERTIES:

The following are the properties of Hydrogen.

- 1. It is very light and can penetrate well into the interior parts.*
- 2. Its density is 7 % of that of air.*
- 3. It has high thermal conductivity (7 times that of air), hence it can dissipate heat at higher rate.*
- 4. Its dielectric strength is 2.5 to 4.5 KV/mm. 5. Liquid hydrogen reduces the resistivity of copper.*
- 6. It is de-oxidizing agent and hence reacts with an oxide coating formed on surfaces to form water.*
- 7. It prevents fire accidents.*
- 8. It is non-toxic and non-metallic at normal temperature.*
- 9. Most of the hydrogen on earth is in molecules such as water and organic compounds because it readily forms covalent bonds with most non-metallic elements.*
- 10. It is highly inflammable and burns with blue flame forming water.*

APPLICATIONS:

The various applications of Hydrogen are as follows.

- 1. Due to its low density, it is used in high speed electrical rotating machines to reduce windage losses.*
- 2. It is used for cooling purpose in electrical machines.*
- 3. It is used as a fuel (rocket fuel).*
- 4. It is used in the manufacturing of fertilizers and filling weather*

balloons.

5. It is used in the preparation of oxy-hydrogen flame which is used in welding due to production of high temperature.

6. It is used to prepare tungsten filaments.

4.SULPHUR-HEXAFLUORIDE (SF)

It is an electro negative gas and has a strong tendency to absorb free electrons. When sulphur is burnt in an atmosphere of fluorine, sulphur hexafluoride is formed. Its molecules have octahedral geometry, containing six fluorine atoms around sulphur atom. It has many advantages as an insulating gas. The dielectric strength increases with increase in pressure. Hence, it must be used under high pressure which needs a sealed tank construction capable of withstanding high pressure.

PROPERTIES:

The following are the properties of Sulphur-hexafluoride (SF₆) gas.

- 1. It has high dielectric about 2 to 3 times that of air.*
- 2. It is colourless, odorless, non-inflammable and non-toxic gas.*
- 3. Its thermal conductivity is about 2 to 3 times higher than air and has better h transfer ability.*
- hear*
- 4. It has superior cooling properties to those of air and nitrogen.*
- 5. It has a high degree of chemical stability at atmospheric pressure and at temperature upto about 100° C.*
- 6. It is an electro-negative gas hence has strong tendency to absorb free electrons. 7. It is costly and sold by weight in tanks and transported in liquid form.*
- 8. At normal temperature, it is non-corrosive.*
- 9. Its density is 5 times higher than air.*
- 10. It is chemically stable and does not react with structural material upto 500° C.*
- 11. Its properties are affected by moisture.*

APPLICATIONS:

The Sulphur-hexafluoride gas has the following applications. 1. It is used as dielectric in transformers and electric switches.

2. It is used in circuit breakers as arc quenching medium i.e. it cools the arc produced between contacts.

3. It is used as insulating medium between live parts of high voltage equipment.

4. SF6 circuit breakers become more popular in high voltage engineering because of better arc quenching ability of SF6 gas.

5. It is used in medical field and aluminium foundry applications as a trace gas in reduction of porosity, corrosive, toxic. 6. It is used as insulating medium in gas-insulated substations and thereby makes the substation compact.

*Time 3 Hours
80*

(Theory 4)

Marks

- 1. What is the effect of porosity?*
- 2. Classification insulating materials on the basis of physical and chemical structures?*
- 3. Glass as insulating material and its use?*
- 4. Impregnated paper, feature and application?*
- 5. What are the types of insulation resistance?*
- 6. What are the properties of SBR?*
- 7. Write the important application of insulating material?*
- 8. Explain mica as an insulating material.*
- 9. Define permittivity*
- 10. Write the uses of glass as an insulator?*
- 11. Write the application of asbestos as insulating material.*
- 12. Explain the use of ceramic and write its properties*
- 13. What are the visual properties of insulation material?*
- 14. What is piezo electric ceramic transducer element?*
- 15. What is A class insulating material?*

- 16. Write the application of wax as insulating material.*
- 17. Classify insulating material*
- 18. What is effect of moisture in insulating material?*
- 19. Explain the reasons for impregnating electric motor winding?*
- 20. Write a short note on insulation resistance?*

Chapter-4

DIELECTRIC MATERIAL

1 INTRODUCTION

The materials which store electrical energy are known as "Dielectric materials". All dielectric materials are essentially insulating materials. The function of an insulating material is to obstruct the flow of electric current through it while the function of dielectric material is to store electrical energy in it. This means the difference between insulating materials and dielectric materials is in their function. The difference between the two types lies in the application to which they are employed. The difference in function of the two types demands that the dielectric materials must have the properties which may not be satisfied by insulating materials. The terms "insulators" and "dielectrics" are often interchangeably. All dielectrics are insulators but all insulators are not dielectrics.

Dielectrics do not possess free electric charges and hence do not conduct electric current. This means that the dielectric materials are usually insulating materials which possess a very low electrical conductivity. In other words the dielectric material provides electrical insulation between conductors. This means dielectric material is a substance that is a poor conductor of electricity, but an efficient supporter of electrostatic field. An electrostatic field can store energy. This property is useful in capacitors, especially at radio frequencies.

The dielectric material can be solid, liquid or gaseous. In practice, most dielectric materials are solids. Examples are porcelain, mica, glass, plastic, Bakelite, paper etc. Some liquids and gases can serve as good dielectric materials. Air, nitrogen, sulphur-hexafluoride are the three most commonly used gaseous dielectrics. Dry air is an excellent dielectric and is used in variable capacitors. Distilled water is a fair dielectric. A vacuum is an exceptionally efficient dielectric. Transformer oil is the most commonly used liquid dielectric.

4.2 DIELECTRIC CONSTANT OF PERMITTIVITY

The property of a dielectric material which gives the information about the storage capacity of electrical energy is known as "permittivity". It plays an important role in dielectrics. It characterizes the tendency of the atomic charge in a dielectric material to distort in the presence of an electric field. The larger the tendency for charge distortion, the larger the value of the permittivity.

Permittivity is a constant of proportionality that relates the electric field in a material to the electric displacement in that material. The permittivity of dielectric material is commonly symbolized by the Greek letter epsilon.

Every medium is supposed to possess two permittivities

(1) Absolute permittivity " ϵ_0 "

(ii) Relative permittivity &

4.2.1 ABSOLUTE PERMITTIVITY (8)

For measuring relative permittivity, vacuum or free space is taken as reference medium. The absolute permittivity of free space or vacuum is 8.854×10^{-12} farad/meter. Its 8.854×10^{-12} farad/meter.

symbol is ϵ_0 . Absolute permittivity, ϵ_0

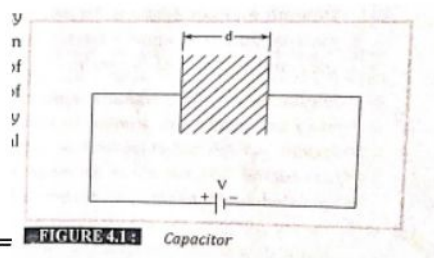
4.2.2 RELATIVE PERMITTIVITY (ϵ_r)

It is also known as 'dielectric constant'. It is defined as the ratio of capacitance of a capacitor having a certain dielectric to the capacitance of a capacitor with air or vacuum as medium. It is represented by the symbol &

If ' C ' is the capacitance of a capacitor with a dielectric medium and ' C_0 ' is the capacitance of the same capacitor in a vacuum, then the relative permittivity is given by

$$\text{Permittivity, } \epsilon_r = C/C_0$$

In engineering applications, permittivity is often expressed in relative rather than in absolute. If so is the permittivity of free space and is the permittivity of the substance, then relative permittivity (dielectric constant) of dielectric material is given by :in engineering applications, permittivity is often expressed in relative rather than in absolute. if so is the permittivity of free space and is the permittivity of the substance, then relative permittivity (dielectric constant) of dielectric material is given by :



Relative permittivity, $\epsilon_r = E =$

The relative permittivity of vacuum is 1.00 and that of air is 1.00058 which is taken as unity. Relative permittivities of dielectrics other than air are more than one. The relative permittivity of some materials are given below in the Table 4.1.

TABLE 4.14 The Relative Permittivities of Different Dielectric Materials

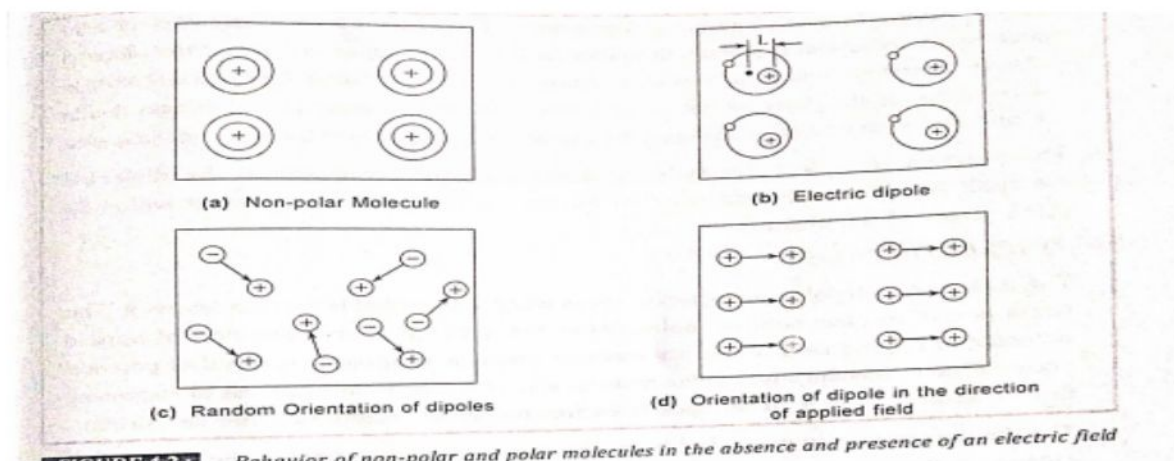
S.No.	Dielectric Material	Relative Permittivity
1.	Air	1.0
2.	Bakelite	5 to 6
3.	Glass	5 to 8
4.	Mica	3 to 8
5.	Paper	2 to 2.5
6.	Porcelain	4 to 7
7.	Transformer oil	2.0

4.3 **POLARISATION**

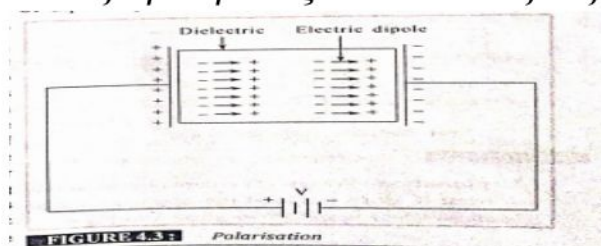
When an external voltage is applied to the dielectric material, the nucleus of the atoms being positively charged is attracted towards the negative side (lower potential) and the electrons being negatively charged are towards the positive side (higher potential) of the supply. Hence, the material gets polarized.

Definition: Polarisation is defined as the definite orientation of dipoles in a dielectric material due to an applied electric field. The degree of polarisation depends upon the dielectric constant of the material.

Explanation: The matter is composed of molecules. The molecules of a dielectric may be polar or non-polar. In non-polar molecules, the centre of the nuclei and electrons orbit coincides as shown in Fig 4.2 (a). When such a molecule is placed in an electric field, the electrons are attracted by the positively charges of the anode and repelled by the negatively charges of the cathode. Due to large attractive force exerted by the nucleus, the electrons do not move away from their atoms, but will undergo slight displacement within the atom itself in the direction of the positively charged electrode. The centers of the nucleus and the electron orbit are displaced by a distance L' as shown in Fig 4.2 (b). This displacement of electrons within an atom due to the effect of electric field is called "electronic polarisation". Due to electronic polarisation, the positive and negative charges in the atom of a dielectric are separated so that it behaves like an electric dipole.



The molecules of many dielectric materials possess an electric dipole movement without having an external electric field. In such molecules, there is a displacement between the positive and negative charges. Such a materials are called polar materials in which the dipoles are oriented in a random fashion in the absence of an electric field as shown in Fig. 4.2 (c). Under the influence of electric field, these dipoles orient themselves in the direction of applied field as shown in Fig. 4.2 (d). The stronger the field, the greater will be the number of dipoles pointing in the direction of the field.

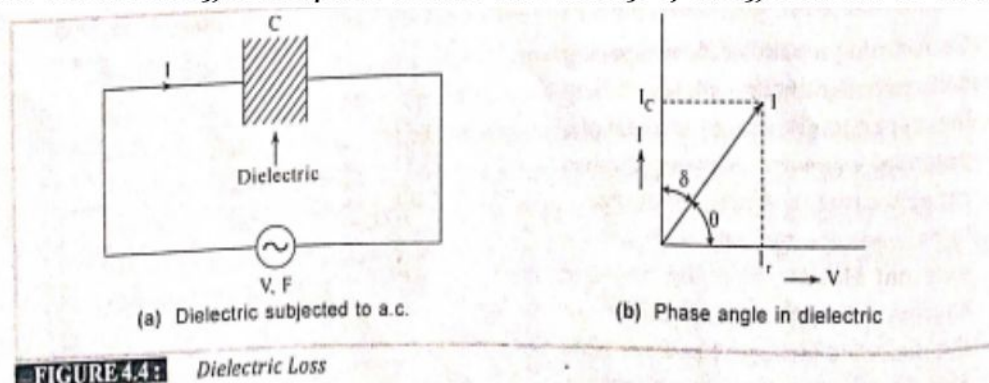


Considering a parallel capacitor with two definite potential will exists between the two capacitor plates. When a dielectric is in between the two plates, due to an external electric field the negative charges will appear on the surface of the dielectric facing the positive plate and the positive charges will appear on the surface of the dielectric facing the negative place of the capacitor as shown in Fig 4.3. Thus the electric field induced within the dielectric due to polarisation will be in a direction opposite to the applied field. The resultant field therefore gets weakened. Hence, the potential difference appearing across the plates is reduced. Therefore, when the plates are connected across a battery, charge will store in the capacitor As the potential difference between the two charged plates reduces due to the introduction of a dielectric material, the charge storing capacity of the capacitor increases. The polarisation effect depends upon the dielectric constant of the dielectric material. A better dielectric material having higher dielectric constant will show more effect of polarization.

4.4 DIELECTRIC LOSS

The dielectric material is subjected to stress when a potential is applied across it. This stress is due to alignment of molecules in the material in the direction of applied potential or electric field. When an alternating voltage is applied, reversal of potential occurs. Due to this the stress in the material also reverses. In this process of alignment, the molecules in the material have to overcome certain amount of internal friction.

The amount of energy expended in this process is wasted as heat in the dielectric. It means when a dielectric material is subjected to alternating voltage, some of the electrical energy is absorbed by the material and this energy is dissipated as heat. This wastage of energy is known as "dielectric loss".



Definition: The loss appearing in the form of heat due to reversal of electric stresses in the process of molecules rearrangement is known as "dielectric loss". **Explanation:** Whenever a dielectric material is subjected to an a.c. voltage, the leakage current 'I' does not lead the applied voltage 'V' by exactly 90°. The phase angle is always less than 90° as shown in Fig 4.4. The complementary angle & 900 is called dielectric loss angle. When a ac voltage 'V' voits at a frequency of Hz is applied to a dielectric material having a capacitance of 'C' farads, the dielectric power loss is given by:

$$P = VI \cos \theta$$

From the Fig 4.4 (b), we have $\theta = 90 - \delta$ and $I = \frac{I_c}{\cos \delta}$

$$\therefore P = V \frac{I_c}{\cos \delta} \cos (90 - \delta)$$

$$= V \frac{I_c}{\cos \delta} \cdot \sin \delta \quad (\because \cos (90 - \delta) = \sin \delta)$$

$$= V I_c \tan \delta \quad \left(\because I_c = \frac{V}{X_c} \right)$$

$$= V \frac{V}{X_c} \tan \delta$$

Where X_c is called capacitive reactance and is equal to $\frac{1}{2\pi f C}$

$$\therefore P = V \frac{V}{\frac{1}{2\pi f C}} \tan \delta$$

$$= V^2 2\pi f C \tan \delta \quad \dots (i)$$

From the above equation, it is seen that power loss dependant on Tan 5 The greater the Tan8. the

larger will be the dielectric loss. Good dielectric materials have $\tan \delta$ as low as 0.001 to 0.0001. The dielectric loss is also proportional to frequency f . At ordinary frequencies the dielectric loss may not be considerable, but at higher frequencies the loss will be more.

4.5 ELECTRIC CONDUCTIVITY OF DIELECTRIC & THEIR BREAKDOWN

All electrical apparatus are designed to operate within a defined range of voltage. If the operating voltage is increased gradually, then at some value of voltage a breakdown will occur and spoiling the insulation property permanently. Definition: The dielectric strength is defined as the voltage at which insulating material will result in the destruction of its insulating properties. It is expressed in KV/mm or KV/cm. It is also known as electric strength.

Applications of Dielectrics: The most common applications of dielectric is as a capacitor to store electrical energy.

There are various types of capacitors available in the market. The capacitors are generally classified according to the dielectric material used in them. The following are the applications of different capacitors.

1. Capacitors using vacuum, air or gases as dielectric are used in radio frequency circuits and in low frequency measuring circuits where precision is highly desirable.
2. Capacitors using mineral oil as dielectric are used as static condenser for power factor improvement of small installations.
3. Capacitors using a combination of solid and liquid dielectric such as oil-impregnated paper capacitors are used for power factor correction in electric power distribution systems and radio circuits.
4. Capacitors with only solid dielectric such as glass, mica, titanium oxide etc are used as standard capacitors in laboratories.
5. Capacitors with insulating oxide film as dielectric (electrolytic capacitors) are used where very large capacitances are required.

4.6 DIELECTRIC PROPERTIES

All insulating materials fail at some level of applied voltage, and 'dielectric strength' is the voltage a material can withstand before breakdown occurs. Dielectric strength is measured through the thickness of the material (taking care to avoid surface effects) and is normally expressed as a voltage gradient (volts per unit length). Note that the voltage gradient at breakdown is much higher for very thin test pieces (<100 μ m thick) than for thicker sections.

The value of dielectric strength for a specimen is also influenced by its temperature and ambient humidity by any voids or foreign materials in the specimen, and by the conditions of test, so that is often difficult to compare data from different sources.

Test variables include electrode configuration and specimen geometry, and the frequency and rate of

application of the test voltage. Standard strategies include:

The 'short time' test, increasing the voltage from zero at a predetermined rate (usually between 100 to 3,000v/sec) until breakdown occurs.

The 'step-by-step' test initially applying half the short time breakdown voltage and then increasing this in equal increments, holding each level for a set period of time.

4.7 APPLICATION OF DIELECTRICS

4.9 The most common application of dielectric is as a capacitor to store energy. Capacitors are classified according to the dielectric used in their manufacture. They can be broadly classified under the following categories:

1. Capacitor using vacuum, air or gases as dielectrics.

2. Capacitor using mineral oil as dielectric.

3. Capacitor using a combination solid-liquid dielectric. 4. Capacitors with only solid dielectric like glass, mica etc. The first type is used in applications where energy loss in the capacitor should be small and the value of the capacitance is also limited. It is essentially used in high-frequency circuits like radios and for frequency measuring devices when precision is important. Oil-filled capacitors are used in applications where the capacitance required is large and a fair degree of power loss is tolerated. It is mainly used at static condenser for power-factor (PF) improvement of small installations.

Oil-impregnated paper capacitors are used to make capacitors of large value and small size. Precision is not important in these applications. They too find application in PF improved devices and radio circuits on account of their cheapness and versatility.

Mica, a solid dielectric, is used to make standard capacitors for laboratories. Its dielectric constant does not change much with temperature and time (it does not age). It has high dielectric constant and insulation resistance and low dielectric loss. It is the ideal dielectric known today.

THEORY - 4

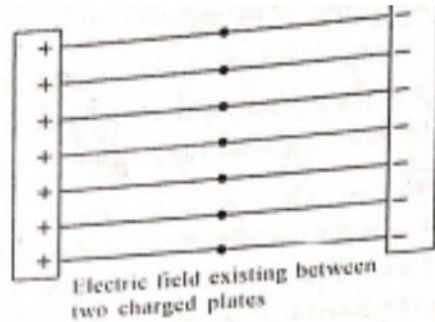
Time-3hr.

Mark-80

1. Define dielectric loss?

2. What is permittivity?

3. Write short notes on polarization?



4. Define a dielectric material.

5. Write a short note on dielectric constant?

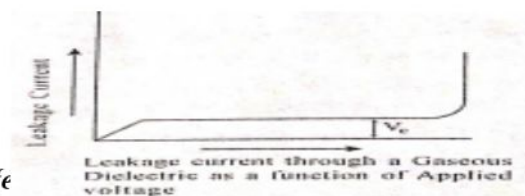
6. What is dielectric flux density?

7. What is polarization?

8. Explain dielectric loss?

9. What is ionization by collision?

10. Explain leakage current through a gaseous dielectric.



11. What is the effect of dielectric loss?

12. Explain the application of dielectric?

CHAPTER -5

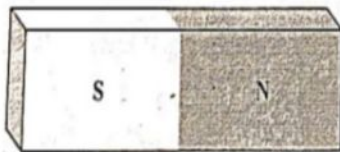
MAGNETIC MATERIALS

5.0 INTRODUCTION

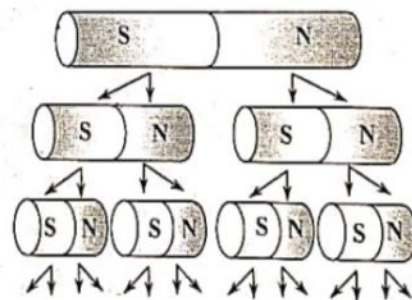
The materials which can be magnetised and are attracted by a magnet are called "**Magnetic materials**". The magnetic materials create a magnetic field around them in which energy transformation is taken place. They provide a path for the magnetic flux, iron, steel, nickel, cobalt etc are some of the most commonly used magnetic materials. When a current flows through a coil, it creates a magnetic motive force (m.m.f). In most materials the direction of motion of the electrons in various orbits is such that they develop m.m.f's in opposite directions and thus cancelling each other. However, in magnetic materials such as iron, steel, nickel, cobalt there are number of un-neutralized orbits, which produce a magnetic field.

Magnetic materials play an important role in the field of electrical engineering. They are used for making magnetic circuits in electromagnets, machines, relays, transformers etc. In all rotating machines, the electromechanical energy conversion is taking place through the magnetic field. The energy storing capacity of the magnetic field is nearly 25,000 times greater than that of electric field.

Magnets have two opposite kinds of magnetic poles namely north pole 'N' and south pole 'S' as shown in Fig. 5.1 (a). Like poles (S-S or N-N) repel each other but unlike poles (N-S) attract each other. Magnetic poles always available in pairs, there is no isolated single pole. If a bar of magnet is broken into two parts, two new poles would appear at the breakage point and thus each part would become a complete magnet itself as shown in Fig. 5.1 (b).



(a) Bar Magnet



(b) Breaking of Magnet

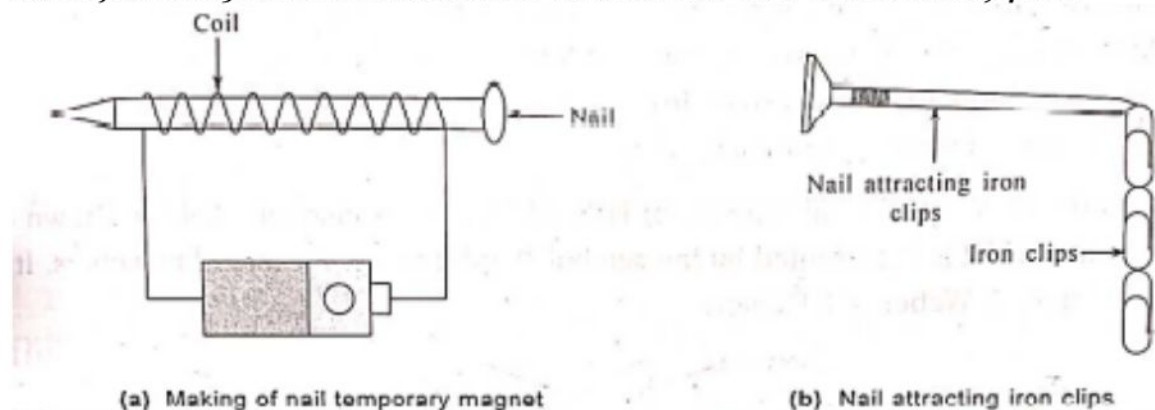
FIGURE 5.1*Properties of Magnet*

Magnets may be natural or artificial. Natural magnets are formed from the iron ores and are obtained from mines which are comparatively weak and hence have no practical uses. Artificial magnets are made from iron, nickel, cobalt, steel or alloy materials and used in all electrical machines and equipments. The magnets may be classified into permanent or temporary magnets depending on their ability to retain

magnetism. Permanent magnets retain their magnetism for a long time and have high retentivity power. These are mainly prepared from steel which is generally much harder than soft iron. These are extensively used in small d.c. motors, electrical measuring Instruments, telephone receivers, speedometers etc.

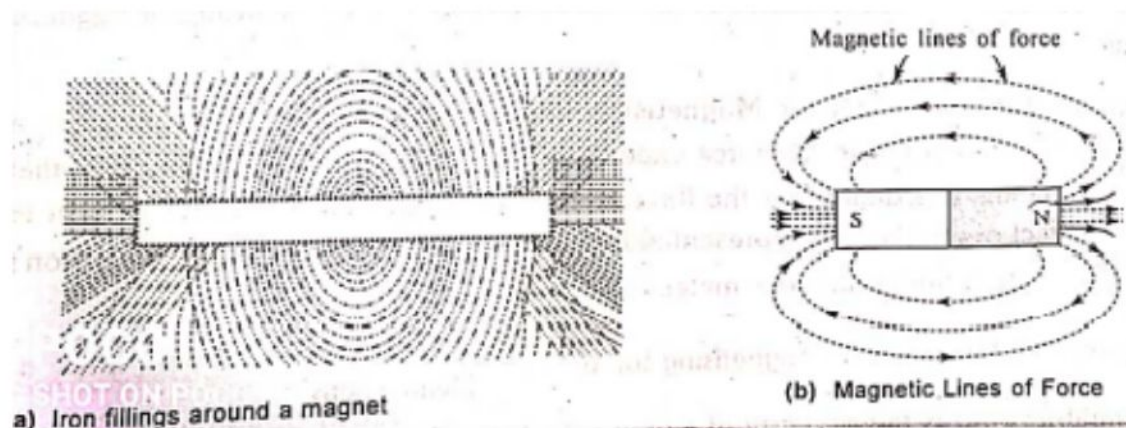
FIGURE 5.2*Making of Temporary magnet by Passing Current*

Temporary magnets are made temporary by passing direct current through a coil wound over iron or steel pieces as shown in Fig 5.2. These magnets lose most of their magnetism when d.c. source removed and have low retentivity power.

**5.1 TERMS ASSOCIATED WITH MAGNETIC MATERIALS**

Before proceeding further, it is of great importance to know the various terms to understand the magnetic behavior of a material. The following are some of the terms related to magnetic material.

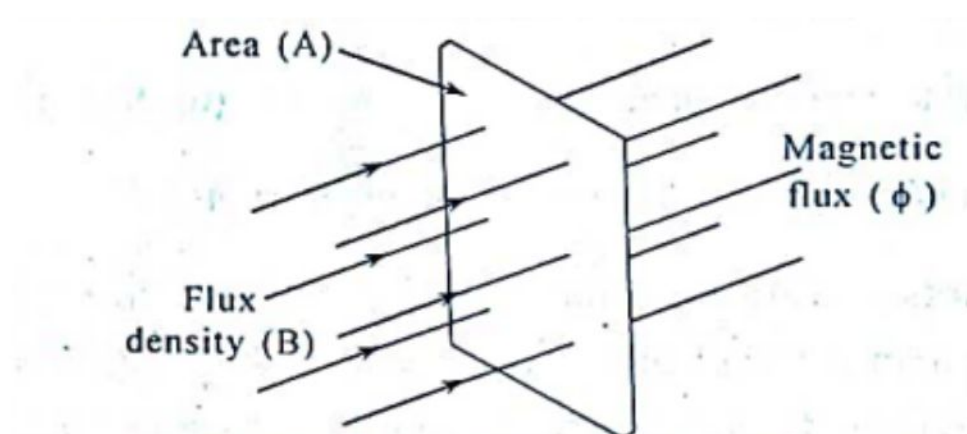
1. Magnetic Field : The space or region of area around the magnetic material within which magnetic effects can be observed is known as magnetic field. The magnetic field is represented by magnetic lines of force.



The magnetic field surrounding a bar magnet can be seen in the magnetograph as shown in Fig 5.3 (a). A magnetograph can be created by placing a piece of paper over a magnet and sprinkling the paper with iron filings. The iron particles align themselves with the lines of magnetic force produced by the magnet.

2. Magnetic Lines of Force: The magnetic field around a magnet is invisible, hence the field is represented by imaginary lines of force as shown in Fig. 5.3 (b). These lines form closed loops and always start from North Pole and end at South Pole. The lines of force never intersect with each other.

3. Magnetic Flux: The total number of lines of force in a magnetic field is known as magnetic flux. It is represented by the symbol (ϕ) and is measured in webers. It is proved that, 1 Weber = 10 lines.



FIGURES.5.4 : Representation of Magnetic Flux and Flux Density

4 Magnetic Flux Density: The magnetic flux density, at any point is defined as the magnetic flux per unit area at that point. It is denoted by the letter 'B' and is measured in webers/m² or Tesla.

If " Φ " webers flux is passing through an area of ' A ' m^2 , then the flux density,

$$B = \Phi / A$$

6. Magnetic Field Strength or Magnetising Force Field strength at any point in a magnetic field is equal to the force experienced by a unit North Pole placed at that point. It means it is defined as the force exerted by a magnet on another magnet to either attract or repel it. It is represented by the letter ' H ' and is measured in Newton's per Weber or Ampere-turn per meter.

m. m. f

Magnetic field strength or magnetising force, $H = \frac{\text{m. m. f}}{\text{Mean length}}$

7. Permeability : It is the property of a material which allows flux freely through it. It is represented by the letter μ (mu) and is measured in Henry/meter. Permeability value varies from material to material depending upon the material and temperature. A good magnetic material should possess high permeability.

$$\therefore \text{Permeability, } \mu = \frac{\text{Flux density } B}{\text{Magnetising Force } H} = \frac{B}{H} \text{ Henry/meter}$$

8. Residual Magnetism: The magnetic flux density which still remains in a magnetic material even when the magnetising force is completely removed is known as residual magnetism.

9. Retentivity: It is the ability of a magnetic material to retain magnetism in it even after the removal of magnetising force. Steel possesses this property while soft iron does not possess it.

10. Reluctance: It is the property of a magnetic material which opposes the magnetic flux through it. It is denoted by the letter ' S ' and is measured in ampere-turn/Weber.

Reluctance, $S = \frac{l}{\mu_0 \mu_r A}$

Where l = length of magnetic path in meters
 A = cross-sectional area of magnetic material in meters
 μ_0 = Permeability of free space
 μ_r = Relative permeability

The value of permeability for free space or air μ_0 is $4\pi \times 10^{-7}$ Henry/meter. The relative permeability μ_r varies from material to material and depends on the degree to which the material is capable of being magnetised. It may have a value as high as 2500.

11. Magnetization (M): It is defined as a measure of increase in magnetic flux due to the insertion of a given material into a magnetic field strength 'H'.

12. Susceptibility : It is defined as the ratio of magnetization 'M' to the magnetising force 'H'. It is dimensionless quantity.

$$\text{Susceptibility, } \chi = \frac{\text{Magnetisation}}{\text{Magnetising force}} = \frac{M}{H}$$

5.2 CLASSIFICATION OF MAGNETIC MATERIALS

The magnetic properties of the materials are characterized by the relative permeabilities. According to their relative permeabilities the magnetic materials are classified into three groups namely:

1. Ferromagnetic materials
2. Paramagnetic materials
3. Diamagnetic materials

5.2.1 DIAMAGNETIC MATERIALS

Diamagnetic Materials: The relative permeabilities of these materials are slightly less than unity. These materials are slightly magnetised when placed in a strong magnetic field and act in the direction opposite to that of applied magnetic field. They are repelled by a magnet. They have a weak, negative susceptibility to magnetic fields. They do not retain the magnetic properties when the external field is removed.

Ex: Silver, Copper, Gold, Bismuth, Zinc, Mercury, Lead, Sulphur, Wood etc.

5.2.2 PARAMAGNETIC MATERIALS

Paramagnetic Materials: The relative permeabilities of these materials are slightly greater than unity. These materials are slightly magnetised when placed in a strong magnetic field and act in the direction of the magnetic field. They are weakly attracted by a magnet. The relative permeability is small but positive. They have a small, positive susceptibility to magnetic fields. They do not retain the magnetic properties when the external field is removed.

Ex: Aluminium, Platinum, Tin, Magnesium, Manganese, Lithium, Tantalum etc.

5.2.3 FERROMAGNETIC MATERIALS

Ferromagnetic Materials: The relative permeabilities of these materials are much greater than unity and are dependent on field strengths. These materials are strongly attracted by a magnet. The relative permeabilities varying from

several hundreds to many thousands. They have a large, positive susceptibility to magnetic fields. They are able to retain their magnetic properties after the external field has been removed.

Ex: Iron, Nickel, Cobalt, Steel and some of their alloys.

5.3 MAGNETISATION CURVE OR B-H CURVE

The curve which shows the relation between the flux density ' B ' and the

magnetising force ' H ' of a ferromagnetic material is known as "Magnetisation curve or B-H curve". In a ferromagnetic material, the flux density ' B ' increases when the external magnetising force ' H ' is increased. Fig 5.5 shows the general shape of B-H curve of a ferromagnetic material. In general it has four regions, OA, AB, BC and CD. If the external

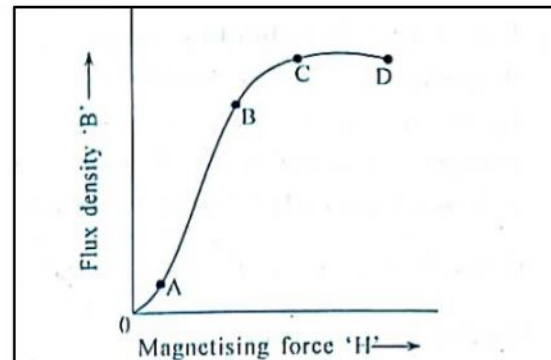


FIGURE 5.5: Magnetisation Curve of a Typical Ferromagnetic material

magnetising force ' H ' is zero, the flux density ' B ' also zero, hence the curve starts from zero. In the beginning during the region OA the increase in flux density is very small with magnetising force. In the region AB, the flux density increases almost linearly with magnetising force and hence the curve is almost straight line. In the region BC the increase in flux density is again small and in the region beyond point 'C', the flux density is almost constant. After point 'C' the flux density will not increase further with the corresponding increase in the magnetising force ' H '. This happens when the material becomes saturated with flux and the curve CD is parallel to X-axis. The flat part CD of B-H curve corresponds to magnetic saturation of the material and the point 'C' is called **saturation point**.

The shape of B-H curve depends on the type of material. Fig 5.6 show the B-H curves of different magnetic materials.

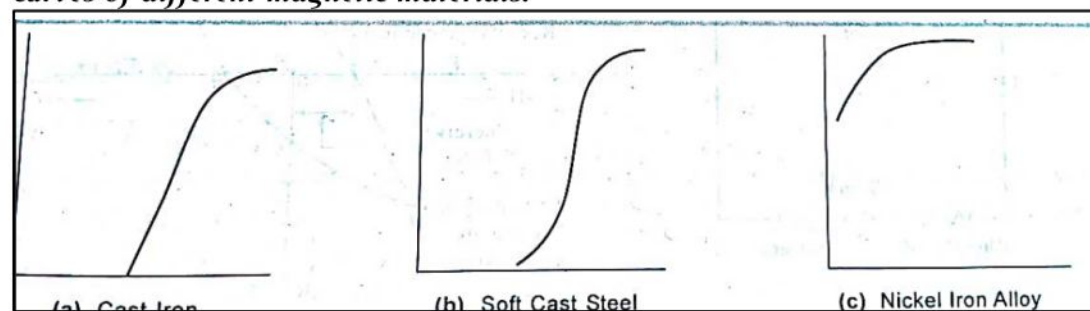


FIGURE 5.6: Magnetisation curves of different magnetic materials

HYSTERESIS LOOP

When a magnetic material is magnetised in a strong magnetic field, it retain a considerable amount of magnetism after the removal of magnetising force. The word SHOT ON PALT '*hysteresis*' means '*lagging behind*'. Hence, the phenomenon of lagging of magnetic flux density '*B*' behind the magnetising force "*H*" is called "*hysteresis*". If a completely demagnetised ferromagnetic material is magnetised through a cycle of magnetisation by increasing the magnetising force *H* from zero to maximum, then decreased to zero, the variation of flux density '*B*' with magnetising force *H* will be represented by closed loop called "*hysteresis loop*" as shown in Fig. 5.7 (b) by ABCDEFA..

Considering an iron piece which is wound with a number of turns of wire as shown in

Fig 5.7 (a). The magnetising force 'H' $\left(H = \frac{NI}{l} \right)$ can increased or decreased

by increasing or decreasing the current I through the wire. When the switch 'S' is in open position, no current will flow through the wire and hence magnetising force is zero. If the magnetising force is increased from zero to a certain maximum, the flux density " B " (BH) increases from zero to maximum represented by curve OA. When saturation arrives, the increase in flux density almost ceases even though magnetising force may be increased. If now the magnetising force is gradually reduced to zero it is found that the original curve OA is not retraced. At ' H ' equal to zero the material is still magnetised and the flux density has the value OB and is called as residual magnetism or remanance. The power of retaining the residual magnetism is called the retentivity of the material.

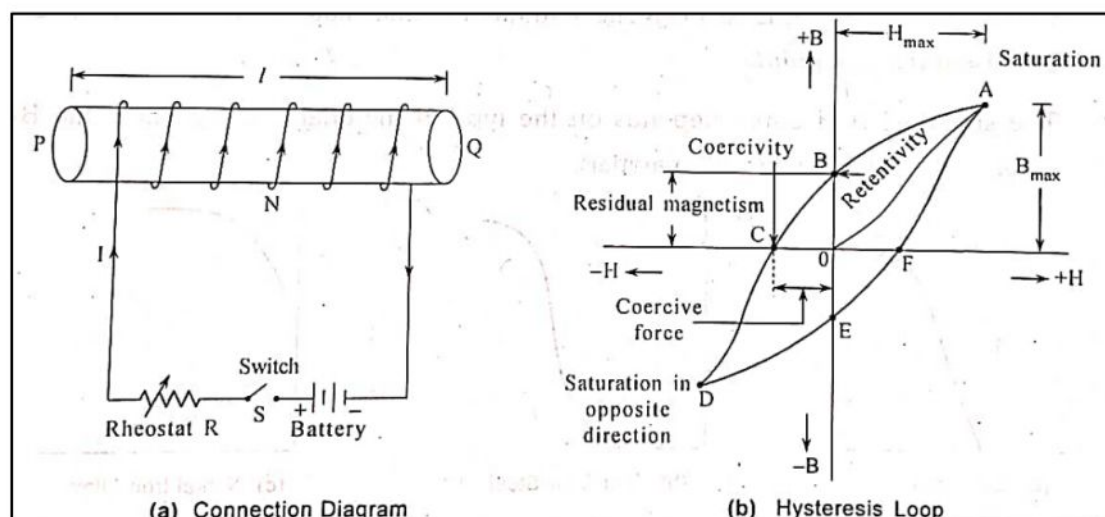


FIGURE 5.7 *Hysteresis Loop*

Now, if the direction of flow of current is reversed, the magnetising force " H " also reverses. In order to demagnetise the material completely, the magnetising force must be reversed. If the current is increased in reverse direction until the flux density becomes zero, at the instant where ' B ' is zero, the magnetising force ' H ' is OC and is called coercive force. Hence, the force required to neutral residual magnetism is known as coercive force. The increase in H in the reverse direction will increase B in the reverse direction and again at D saturation occurs. If the demagnetizing force H is gradually reduced to zero, the curve DE is obtained. When $H = 0$, $B = OE$, OE is the residual magnetism in reverse direction and to neutralize this, H must be increased in positive direction. Further increase of H in the positive direction, saturation will occur at point A and the curve EFA is obtained. Thus a closed loop $ABCDEFA$ is obtained which is usually known as 'hysteresis loop or magnetic cycle'.

From hysteresis loop it is noted that, B lags behind H and the two never attain zero value simultaneously. The shape of the hysteresis loop depends upon the type of ferromagnetic material. The hysteresis loops for different types of materials are shown in Fig 5.8.

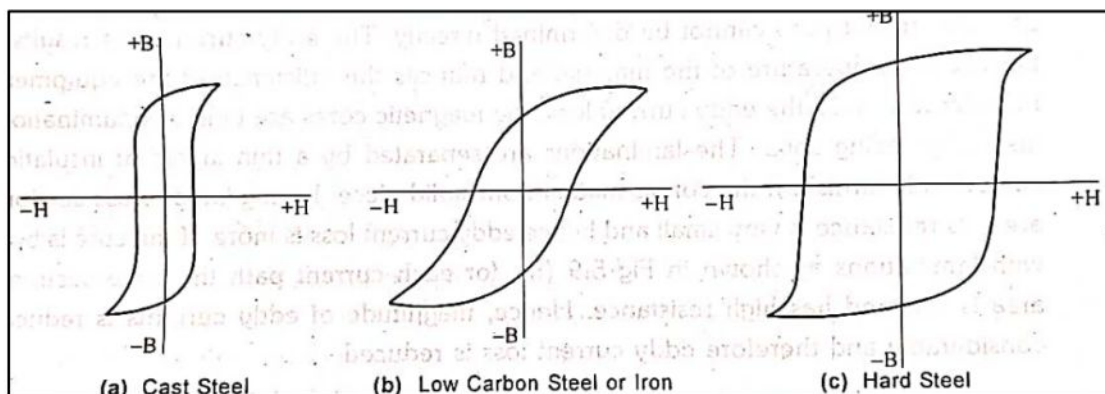


FIGURE 5.8: *Hysteresis Loops for different materials*

HYSTERESIS LOSS

When a ferromagnetic material is subjected to repeated cycles of magnetisation, it results in power loss in the form of heat and is known as "hysteresis loss".

Hysteresis loss is equal to the energy consumed in magnetising and demagnetizing a magnetic material. It is due to the friction between the molecules of the material as they are shafted in direction by the magnetising force. It is proportional to:

- (i) Area enclosed in the hysteresis loop
- (ii) of the alternating
- (iii) Volume of material

Hysteresis loss is significant in determining the heating, rating and efficiency of rotating machines, transformers and a.c. operated magnetic devices. Therefore, magnetic cores of silicon steel which has low hysteresis co-efficient.

EDDY CURRENT LOSS

When a magnetic material is subjected to an alternating magnetic flux, according to the Faraday's laws of electromagnetic induction an e.m.f is induced in the cores. The em.f causes an electric current in the body of the core material due to its small resistance. This current is known as "**eddy current**". The power loss due to eddy currents is known as "**eddy current loss**"

The power loss due to eddy currents is proportional to $1/R$, where R is the magnitude of eddy current and R is the resistance of eddy current path provided by the material. But, it is difficult to determine it because the magnitude of eddy currents and resistance of eddy current path cannot be determined directly. The eddy current loss results in the rise of temperature of the material and reduces the efficiency of the equipment. In order to reduce the eddy current loss, the magnetic cores are built with laminations instead of being solid. The laminations are separated by thin layers of insulation coated with varnish. If the core is made of one solid piece, having larger cross-sectional area, its resistance is very small and hence eddy current loss is more. If the core is built with laminations as shown in Fig 5.9 (b), for each current path the cross-sectional area is less and has high resistance. Hence, magnitude of eddy currents is reduced considerably and therefore eddy current loss is reduced.

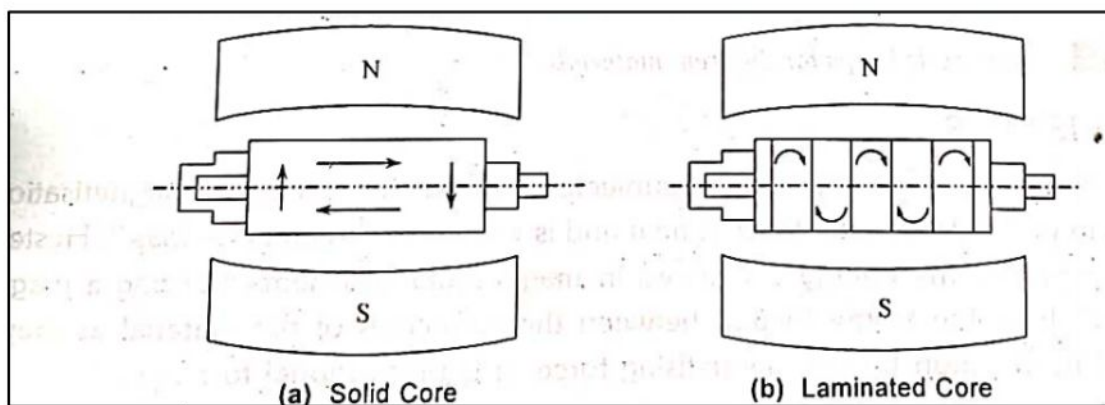


FIGURE 5.9

It is found experimentally that the eddy current loss is proportional to square of the maximum flux density (B), square of the frequency (f) and is given by the equation,

Eddy current loss,

$$W_e = KB_{\max}^2 f^2 t^2 V \text{ watts}$$

Where,

K = Eddy current coefficient and depends on the type of the magnetic material = $1/\rho$ i.e., K is inversely proportional to the resistivity of the material

$B_{\max}$ = Maximum flux density in wb/m^2

f = Frequency of a.c. supply in Hertz

t = Thickness of laminations in meters

V = Volume of the core material in m^3

5.6 CURIE POINT

When a magnetic material is heated, its molecules vibrate more violently. If the temperature increased further individual molecular magnets get out of alignment and the magnetic strength of the material is reduced. At particular (critical) temperature magnetic material lose its properties partially or even completely.

Curie point is the temperature at which ferromagnetic material lose their magnetic properties. This temperature is a characteristic of magnetic material and differs from material to material. At its curie temperature iron ceases to be ferromagnetic and becomes paramagnetic.

5.7 MAGNETOSTRICTION

When a ferromagnetic material is magnetised, a small change in dimensions of the material takes place. There is a small extension of length with corresponding reduction of cross-section area of the material. When the material is subjected to rapidly alternating magnetic field, there is a rapid and continuous extension and contraction of the material. This is called "Magnetostriction". This property is mostly found in nickel and nickel iron alloy. Magnetostriction is the major cause of hum in transformers and chokes.

5.8 SOFT AND HARD MAGNETIC MATERIALS

Ferromagnetic materials are classified into two types namely.

- 1. Soft magnetic material*
- 2. Hard magnetic material*

5.8.1 SOFT MAGNETIC MATERIAL

Soft magnetic materials: *Materials which have a steeply rising magnetisation curve, 1ty small and narrow hysteresis loop are called 'soft magnetic materials'. These have small energy losses during cyclic magnetisation. These materials do not retain their magnetism for any appreciable time after removing the magnetising force. It is easy to magnetise a soft magnetic material. The most widely used soft materials are soft iron, silicon steel, nickel iron alloys and soft ferrites. Fig. 5.10 (a) shown a typical narrow hysteresis loop marked ABCDEFA for soft magnetic materials.*

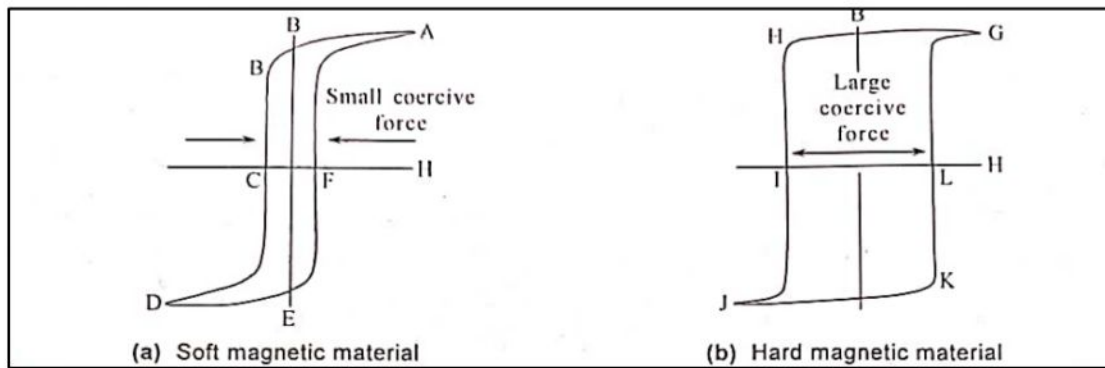


FIGURE 5.10:
material

Typical Hysteresis Loops for Soft and Hard magnetic

PROPERTIES :

The following are the properties of soft magnetic materials.

- (i) They have high permeability.*
- (ii) They have low hysteresis and eddy current loss.*
- (iii) They can be magnetised and demagnetised easily.*
- (iv) They have low residual magnetism (remanance).*
- (v) They have low coercive force.*
- (vi) Large magnetising force is required to induce magnetism.*
- (vii) They have low retentivity.*

APPLICATIONS:

The following are the applications of soft magnetic materials.

- (i) They are used for the construction of core plates of transformers, electrical machines.*
- (ii) They are used to make electro-magnets for electrical machines.*
- (iv) They are used in industries, telephone receivers, relays etc.*

5.8.2 HARD MAGNETIC MATERIAL

Hard magnetic materials: *The materials which have gradually rising magnetisation curve, large and wide hysteresis loop are called **hard magnetic***

materials'. These have high energy losses during cyclic magnetisation. These materials retain their magnetism for a long time after removing the magnetising force. It is difficult to magnetise a hard magnetic material. The most widely used materials are carbon, steel, tungsten steel, cobalt steel, alnico, hard ferrites etc. Fig 5.10 (b) shows a typical broad hysteresis loop marked GHIJKL.G for hard magnetic material.

PROPERTIES:

The following are the properties of hard magnetic materials.

- (1) They have low permeability.*
- (ii) They have high hysteresis and eddy current loss.*
- (iii) It is difficult to magnetise.*
- (iv) They have high residual magnetism (remanance).*
- (v) They have high coercive force.*
- (vi) A high magnetising force is required to attain magnetic saturation of the material.*
- (vii) They have high retentivity.*
- (viii) The hysteresis loop is rectangular in shape.*
- (ix) They have strong magnetic reluctance.*

APPLICATIONS:

The following are the applications of hard magnetic materials.

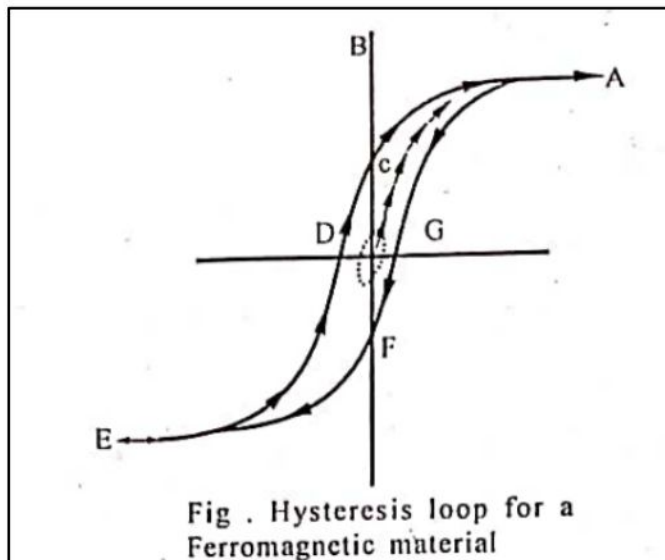
- (i) They are used for making permanent magnets which are used in electrical machines.*
- (ii) They are used in automobiles for steering, wiper, fan motors etc.*
- (iii) They are used in electrical measuring instruments such as galvanometer, ammeter, voltmeter, wattmeter etc.*
- (iv) They are used in loud speakers of audio systems, light weight earphones, hearing aids, computer peripherals etc.*
- (v) They are used in telephones, clocks etc.*

(Theory)

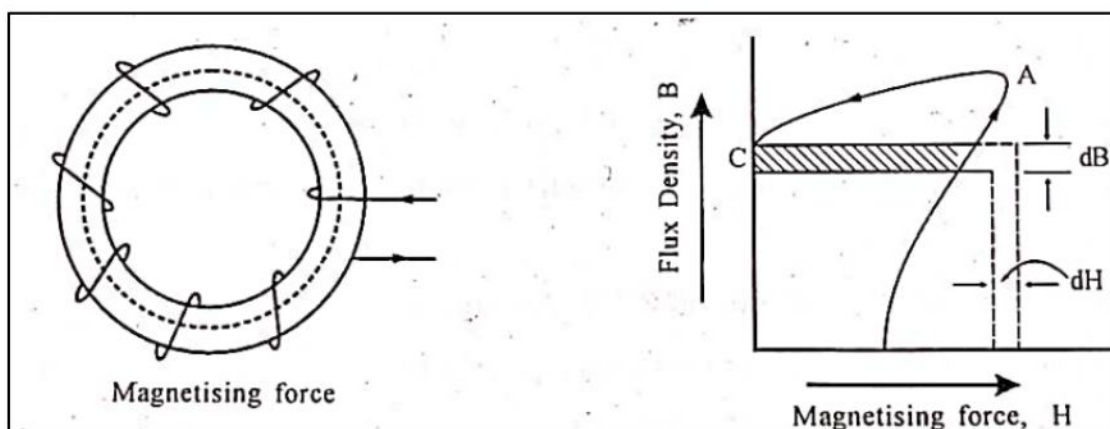
Time - 3hr

mark - 80

1. What is the major cause of hum in transformers and chokes?
2. Define Hysteresis?



3. What is paramagnetic material? Give example ?
3. Write a short notes on soft magnetic materials.
4. Explain energy expended in a magnetic cycle.



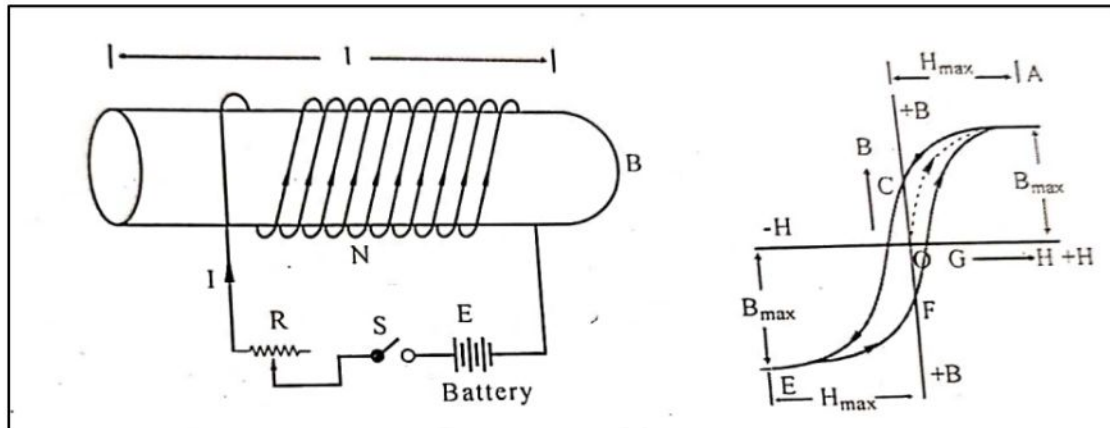
6. What is diamagnetic material? Give example.
7. Write a short note on hard magnetic material.

8. What is ferromagnetic material? Give example.

9. What is spontaneous magnetisation?

10. Explain Steinmetz Hysteresis law?

10. Explain magnetic hysteresis?



12. Write the uses of soft magnetic material.

13. Write the uses of hard magnetic material.

14. Write the uses of soft magnetic material.

CHAPTER-6

MATERIALS FOR SPECIAL PURPOSE

6.1 INTRODUCTION

MATERIALS

ELECTRICAL ENGINEERING

*Electrical engineering is a vast field in which a large number of materials are used for different functions. The materials can be broadly classified as **primary materials** and **secondary materials**. The materials which perform primary functions are called **primary materials** whereas the materials which perform secondary functions are known as **secondary materials**. It is often necessary to use certain materials not directly connected with the given function but to assist in performing the function effectively and efficiently.*

*For example, the structural materials like poles and towers do not perform its function as a conductor for transmission and distribution of power but support the overhead conductors. Similarly, frames of electrical machines and equipments, lead sheathing and steel armouring in cables, conduits in conduit wiring etc. These materials do not perform the primary function of the particular equipment, they perform secondary function. But, without the use of secondary materials the primary function would not be effectively and efficiently. It means secondary materials are used for special purpose and hence they are known as "**special purpose materials**".*

In electrical engineering special purpose materials are classified into six main groups as follows:

- 1. Structural materials*
- 2. Protective materials*
- 3. Thermo-couple materials*
- 4. Bi-metals (Thermostats)*
- 5. Soldering materials*
- 6. Fusing materials*

6.2 STRUCTURAL MATERIALS

*The materials which are used for construction of structures are known as "**structural materials**". The various structures used in electrical engineering are poles, towers, supports for electrical equipments, foundations for rotating machines etc. The most commonly used materials for electrical engineering structures are cast iron, steel, wood, cement, bricks, reinforcement concrete etc.*

Cast iron and steel are widely used for the transmission poles and towers, frames of electrical machines, tanks of transformers and a large number of other applications. Wood is naturally available and is used for poles in distribution of electrical power. Cement, bricks, reinforcement concrete are used in foundations for electrical machines.

6.3 PROTECTIVE MATERIALS

The materials which are used for protection of various materials and equipments are known as "protective materials". In electrical engineering every material and equipment needs protection against atmosphere conditions. The most commonly used protective materials are:

- | | | |
|-----------------|---------------------------------|-----------------|
| <i>1. Lead.</i> | <i>2. Paints and varnishes.</i> | <i>3. Steel</i> |
| <i>tapes</i> | | |

6.3.1 LEAD

Lead: *Lead is soft, heavy and bluish-green metal. It has good malleable and ductile properties. It is high resistant to many chemical actions. It is mechanically weak. It may be used as a pure metal, alloyed with other metals or as chemical compound. In electrical engineering lead is used as plates or grids in storage batteries, as sheathing of high voltage power cables, for making solder and bearings and antifriction metals. Lead sheets are used for sound insulation purpose. Lead forms with many metals and are used for industrial purpose.*

6.3.2 PAINTS AND VARNISHES

Paints and Varnishes: *Paint is a liquid which after application to an object, is converted to a solid film. It is most commonly used to protect, colour or provide texture to objects. It is used to protect the equipment from corrosion.*

Varnish is a transparent, hard protective finish used in wood finishing and also for other materials. It is also used to prevent corrosion of metals.

6.4.1 THERMO COUPLE MATERIALS

Thermo couples can be used for the measurement of temperature. Depending on the range of temperature to be measured, proper materials are to be selected for a thermocouple.

*When a couple of different metals or alloys are joined at their ends and the junction ends are maintained at different temperatures then an e.m.f. exists across the junctions. This e.m.f causes a current to flow from hot junction to cold junction. The e.m.f depends upon the type of metals or alloys and on the difference of temperature between the junctions. As the e.m.f and hence currents induced in the couple are due to the difference of temperature between the junctions, the combination is known as a "**thermo couple**". The e.m.f is called thermo e.m.f and the resulting current is known as thermo electric current.*

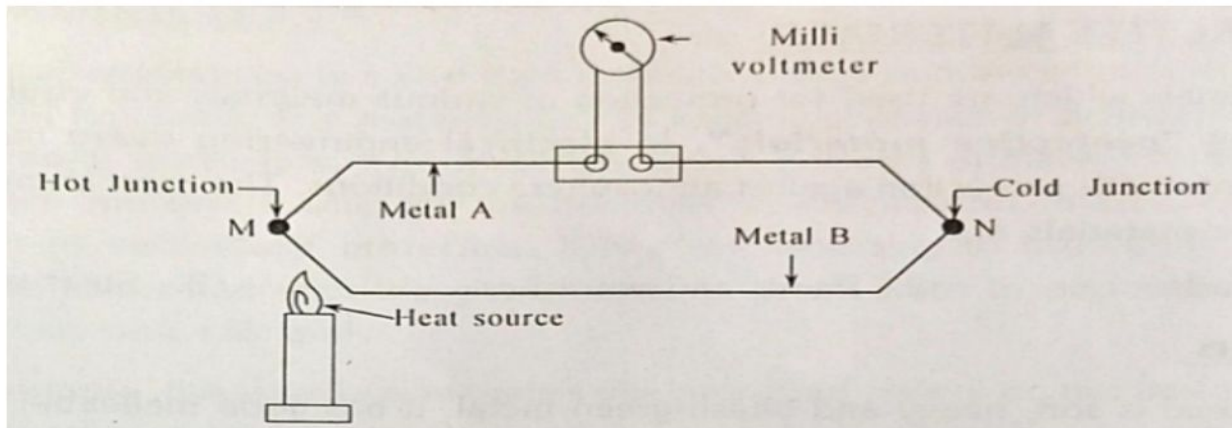


FIGURE 6.1: *Thermo Couple Circuit*

The e.m.f produced by a thermo couple is very small and is measured by a sensitive moving coil millivoltmeter as shown in Fig. 6.1. The millivoltmeter can be calibrated in terms of temperature. If the cold junction N is maintained at constant temperature and the temperature of the hot junction is slowly increased, the millivoltmeter indicates a definite potential difference for any given temperature of junction M. The e.m.f causes a current to flow upto a certain maximum temperature. If the temperature of the hot junction is increased beyond the maximum value, then the current starts decreases and becomes zero at a particular temperature. If the temperature of the junction is still increased, then the current reverses and increases.

The principle of thermo couple is pyrometer which is used to determine the temperature of an object's surface. Materials combination for thermo couples are copper/constantan upto 500°C , Iron/constantan upto 900°C , Nickel/nickel chromium upto 1100°C .

6.4.2 BI METALS OR THERMOSTATS

A bimetal or thermostat is based on the theory that metals expand on heating and contract on cooling. It works on the principle of the heating effect of electric current. Its function is the switch ON and OFF the supply automatically at a certain temperature.

A bimetal or thermostat consists of two metallic strips of different metals or alloys with different coefficients of thermal expansion. The strips are welded together lengthwise. At a certain temperature, the strip will bend and operate a switch or a lever of a switch. A bimetal can be heated directly or indirectly. When heated it bends in such a way that metal with higher coefficient of expansion is on the outside of the arc formed and that with smaller coefficient is on the inside. When cooled the element bends in the other direction. Alloys of iron and nickel with low coefficient of thermal expansion are used as one element of the bimetallic strip. The iron, nickel, constantan, brass etc having high coefficient of thermal expansion is used as other element of bimetallic strip. A typical bimetal is shown in Fig. 6.2.

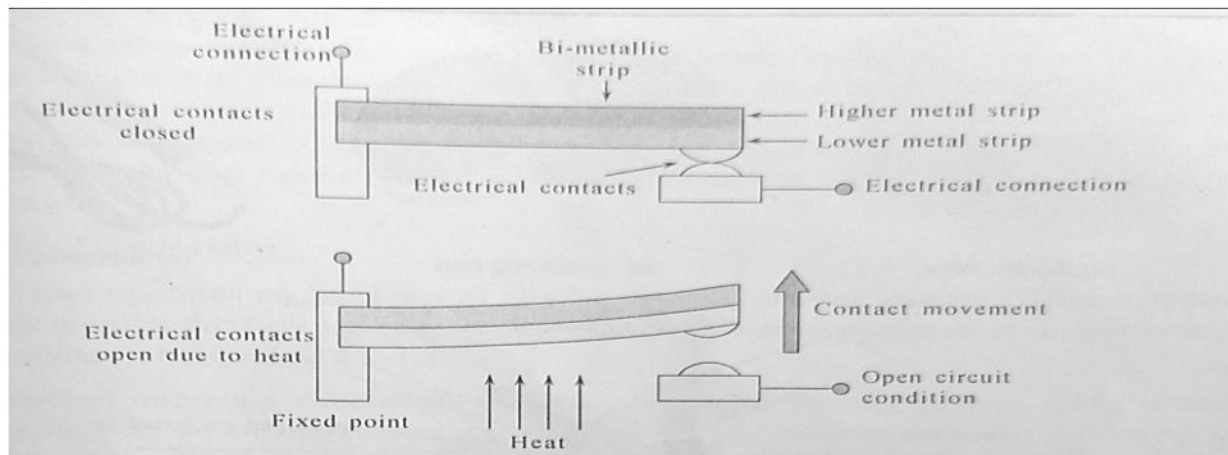


FIGURE 6.2: *Illustration of Bimetallic strip operation*

Bimetallic strips are used in electrical apparatus and devices as relays and regulators. A bimetal relays are used for overload protection of electrical machines and equipments. In order to maintain constant temperature in electric heaters, ovens, furnaces etc a simple bimetallic regulator may be used. Many bimetal devices are adjustable for a particular current. If the current exceeds certain value, the strip will heated enough to bend and break the circuit.

6.4.3 SOLDERING MATERIALS

Solder is an alloy of two or more metals of low melting point which is used to join two or more pieces of metals. The process of joining the pieces of metals is known as "**soldering**". The melting point of solder is lower than that of the materials to be soldered. The most common solder is an ally of 50 % of lead and 50% of tin. Its melting point is about 185° C and its electrical conductivity is about 10 % that of copper. For proper soldering, it is necessary to clean the surface of the metals to make it free from dust, dirt, grease etc and then flux is applied around the joint. The flux is a substance used to dissolve oxides on the surface of metals and to protect against de-oxidation during soldering process. The state of flux may solid or liquid such as zinc chloride, hydrochloride acid (10%) diluted with (90%),Sal ammonia reasin or Tallow.

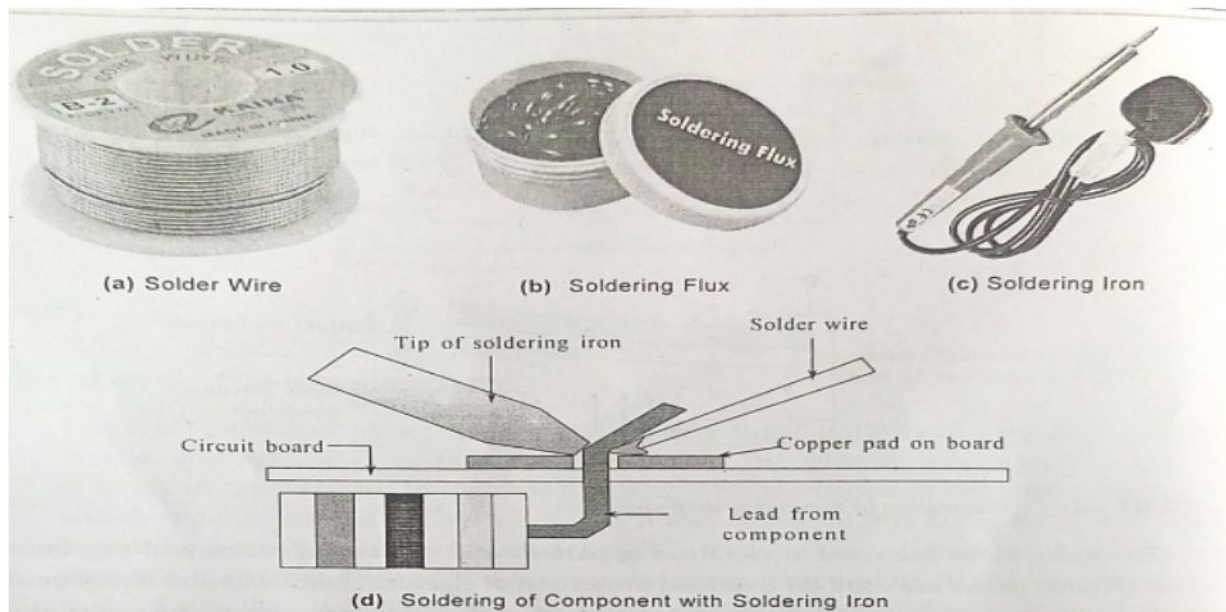


FIGURE 6.3: Soldering Components and Soldering Process

A piece of solder wire is placed over the joint and heated with proper temperature. The solder melts and spreads on the surface of the joint evenly. It is necessary to heat the joint upto a proper temperature. If it is not properly heated, proper soldering cannot be done and if the joint is over-heated the metal is weakened and become brittle. The process of soldering is shown in Fig. 6.3.

Solder used for electrical purpose can be divided into two groups namely soft solders and hard solders.

Soft Solders

Soft solders are composed of lead and tin in various properties. Its melting point is lower than 400°C . These are used for soldering copper, bronze, brass and other such metals. These are used in electronic devices.

Hard Solders

Hard solder is an alloy of copper and zinc. Its melting point is higher than 400°C . These are used for soldering brass, copper, iron, gold, silver, steel etc. These are used in power apparatus for making permanent connections.

6.4.4 FUSE AND FUSING MATERIALS

Fuse is a safety device used for the purpose of protecting an electrical circuit against excessive current. The fuse is a small piece of metal wire connected between two terminals mounted on an insulated base. It is always connected in series with the circuit. In the event of excessive current (due to short circuit or overload), the fuse element melts and opens the circuit thereby protecting the circuit or appliance from damage.

Principle of Operation

The fuse operates on the basis of heating effect of electric current. When a current flows through the fuse element, it gets heated. The heat produced is proportional to the square of the current ($H = I^2 R t$).

In normal operating conditions, the fuse carries the normal current safely without heating. But, when the current exceeds the normal current, the heat produced will increase. If the temperature increases above the melting point of the material of the fuse wire, it melts and breaks the circuit. In this way fuse protects a machine or apparatus or any installation from damage due to excessive current.

Characteristics of Fuse Element Material

The material used for fuse element should possess the following characteristics.

- | | |
|-----------------------|---|
| 1. Low melting point. | 2. High conductivity |
| 3. Low cost. | 4. Free from deterioration to oxidation |

In general, no material possesses all the desirable characteristics.

6.5.3 FUSE MATERIALS

The materials commonly used for fuse elements are:

- | | |
|---------------|--------------------------|
| 1. Lead. | 2. Tin |
| 3. Silver. | 4. Tinned copper |
| 5. Zinc. | 6. Antimony |
| 7. Aluminium. | 8. Alloy of lead and tin |

An alloy of lead (37 %) and tin (63%) is used for small currents i.e. upto 15 A. For currents exceeding 15 A, copper wire fuses are used. The copper is tinned to provide protection against oxidation. Now-a-days silver is extensively used as fuse in spite of high cost as it has all the required characteristics.

Types of Fuses

Fuses may be classified into two types namely:

1. Low voltage type fuses.

2. High voltage type fuses

1. Low Voltage Type Fuses: These may be semi-enclosed rewirable fuses or cartridge type fuses.

Semi-enclosed rewirable fuse is most commonly used in house wiring. It has two main parts namely fuse base and fuse carrier. Fuse base has two terminals to which incoming and outgoing phase wires are connected. Fuse carrier consists of two terminals with strips to which fuse element is connected. These are used upto 100 A. When the fuse element is blown off due to excessive current, the fuse carrier is pulled out and the blown out fuse element is replaced by new one. The parts of rewirable fuse are shown in Fig. 6.4.

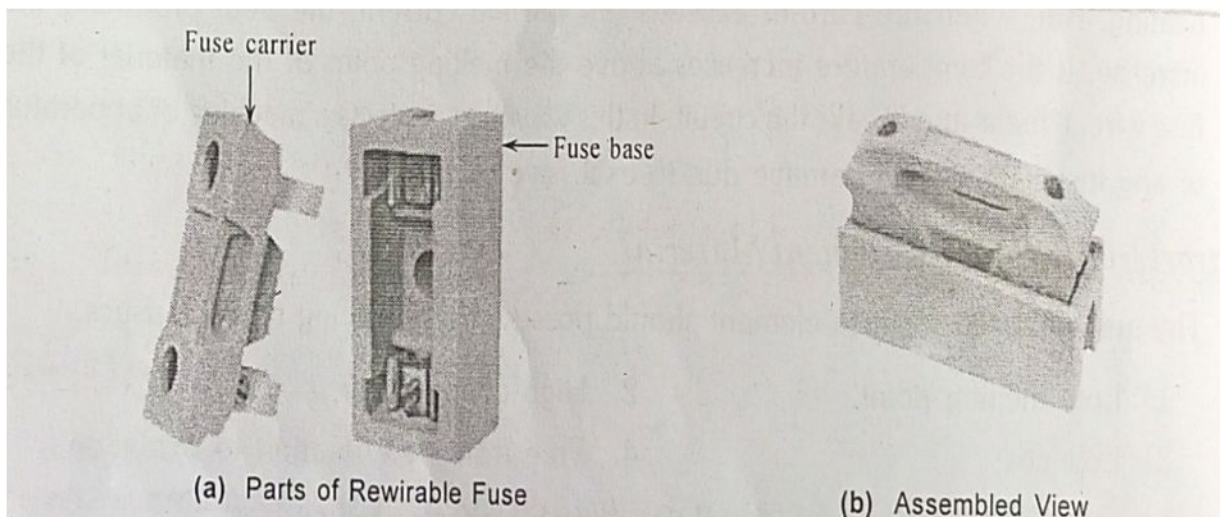
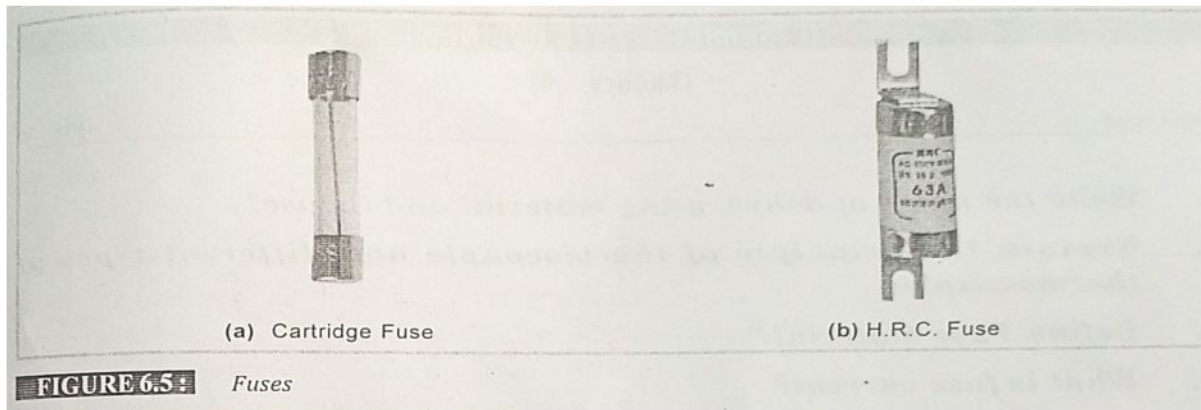


Figure 6.4. Rewirable Fuse

Cartridge fuse consists of a ceramic body and sealed at its ends with a metallic caps. The fuse element generally of silver is fixed between metallic caps. The cartridge is filled with powder material such as sand, calcium, carbonate, quartz etc. On overloads or short-circuits, the fuse element is heated to a high temperature causing it to vaporize. The powder in the fuse cartridge cools and condenses the vapour and quenches the arc at the time of blowing up of fuse. When fuse blows out, it is replaced by a fresh unit as the cartridge cannot be rewired due to its sealing as shown in Fig. 6.5 (a).

2. High Voltage Type Fuses: The fuses used for high voltage are generally High Rupturing Capacity (H.R.C) fuses. H.R.C fuse is a cylindrical in shape and consists of a heat resistant ceramic body having metal end caps. The fuse element generally of silver is fitted between two caps. The space within the body surrounding the element is filled with a powder which acts as arc quenching and cooling medium. The filling material may be chalk, plaster of paris or quartz. These are used upto 66 KV and above. They are available from 1 A to 400 A current rating. A typical H.R.C. fuse of 63 A rating is shown in Fig. 6.5 (b).



6.4.5 DEHYDRATING

Dehydration of natural gas is the removal of the water that is associated with natural gases in vapour form. The natural gas industry has recognised that dehydration is necessary to ensure smooth operation of gas transmission lines. Dehydration prevents the formation of gas hydrate and reduces corrosion. Unless gases are dehydrated liquid water may condense in pipe lines and accumulate at low points along the line, reducing its flow capacity. Several methods have been developed to dehydrate gases on an industrial scale.

The three major methods of dehydration are (1) direct cooling (2) absorption and (3) adsorption molecular.

Usually, the absorption stripping cycle is used for removing large amounts of water, adsorption is used for cryogenic systems to reach low moisture contents.

1. Silical gel (Air dehydrating) breathers are transparent hollow cylinder tubes which contain chemically pure "**Silicium salt**" (Silica gel) with cobalt indicator. This grade silica gel is called as "Indicating grade" of silica gel.

The indicating grade of silica gel, which is filled in the breather is hard blue crystals, which has considerable absorption power of moisture, silica gel absorbs moisture signalling the saturation degree of changing colour follows:

a.) Deep blue silical gel: Completely dry Light blue silical gel: Partly humid

b.) Light blue silica gel : partly humid

c.) Pink silica gel: Saturated with moisture

For proper functioning of transformer, dehumidification of silical gel crystals and removal of dust particle from breathed air is necessary.

6.10 II-/Sem/Elec/2010 (W) (New)

(Theory - 4)

Time 3 Hours

Mark : 80

- 1. Write the name of dehydrating material and its use? Explain the principle of thermocouple and different types of*
- 2 .Explain the principle of thermocouple and different types of thermocouple*
- 3. Define Fuse material?*
- 4. What is fuse current?*
- 5. Write a short note on lead?*
- 6. Write a short note on solders?*
- 7. What is the function of fuse?*
- 8. What is the function of fuse? What are the factors which effect contact performance?*
- 9. Explain the function and uses of thermostat?*
- 10. What are the advantages of fuse?*
- 11. Write a short note on dehydrating material.*
- 12. Write a short note on tape.*